

A Work Project, presented as part of the requirements for the Award of a Master Degree in Finance from the
NOVA – School of Business and Economics.

GreenVolt: The Promising Pipeline
The path to become a pan-European powerhouse

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A Project carried out on the Master in Finance, under the supervision of:

Prof. Gonalo Rocha

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“No matter how sophisticated our choices, how good we are at dominating the odds, randomness will have the last word.”

Nassim N. Taleb

Abstract

This Equity Research report initiates coverage of GreenVolt (GVOLT), which is a Portugal-based developer and independent power producer operating in the biomass, solar PV, wind, and distributed energy generation segments. Through a SOTP, we captured the high visibility and steady cashflows on biomass and distributed generation via DCF and we crystalized solar PV and wind's value via EV/MW, where different technologies and pipeline stages were taken into consideration. We issue a Buy rating, with a Target Price of €9.50/share, implying an upside of c.17%, over the next 12 months. Macroeconomic and industry factors are favourable to GreenVolt's equity story of becoming a powerhouse in the renewable energy generation market. Risks to the upside include in-place biomass remuneration schemes extensions. Risks to the downside include high mortality rate for early-stage projects and management dependency.

Acknowledgement

I would like to thank my Mother, Lúcia, my Father, David, and my Brother, David, for all the love, support, trust, friendship, guidance, wisdom and stability since I can remember. I would also like to thank my boyfriend, Rodrigo, for being the best partner in life.

Additionally, I would like to thank my Equity Research partner, Bárbara Janeiro, for her friendship and companionship. I also thank our supervisor, Professor Gonçalo Rocha, for his feedback and guidance throughout this endeavour.

Keywords: renewables, valuation, generation

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This report is part of the *GreenVolt: The Promising Pipeline* report (annexed), developed by Bárbara Janeiro and Francisca Paulos and should be read as an integral part of it.

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Introduction

This Equity Research report aims at initiating coverage of GreenVolt, which is a Portugal-based developer and independent power producer operating in the biomass, solar PV, wind, and distributed energy generation segments.

To do so, we start by providing a **Company overview**, where we look into GreenVolt's business segments, geographic footprint, value chain, and its recently updated strategy plan for 2022-2026. It follows a **Macroeconomic context**, that allows the reader to understand how do recent worldwide events play a role for a renewable energy generator player like GreenVolt. Then, we provide an **Industry overview**, where we assess demand and supply factors driving renewables, and we present the industry's revenue and costs drivers. Revenue forecast is based on a top-down approach, where we evaluate if GreenVolt has the required portfolio to capture market share (in terms of installed capacity and load factor), if other competitors have the required portfolio to capture market share, and how is the Company affected by long-term trends (in terms of average selling price, regulatory frameworks, and other macroeconomic factors). For the costs' assessment, we forecast feedstock costs, instalment costs, OpEx and CapEx. Then, we analyse some **Risks** that can impact GreenVolt. Having all of these in mind, we are then able to present our **Valuation** case, based on a Sum-of-the-Parts (SOTP) approach, cross-checking with trading multiples and precedent transactions. In this section, we also detail the discount rates used. At the end, this report allows the reader to draw meaningful conclusion on whether to invest or not in GreenVolt.

This present report covers the following sections: Company overview, Macroeconomic context, and Industry overview (demand and supply and revenue drivers). These sections allow the reader to have a deep understanding of the Company and the industry's fundamentals, on which we build on our forecasts and subsequent valuation methods later on. The remaining sections are covered by my Equity Research partner.

This analysis yields a rating of **Buy**, at a Target Price of €9.50/share, implying an upside of c.17%. This is cross-checked with trading multiples (€7.76/share) and past transactions (€11.46/share).

We conclude GreenVolt is well poised to crystalize value in the asset rotation strategy in the near-term, highly enjoying long visibility on biomass assets' cashflows.

Founded: 2002

Headquarters: Porto,
Portugal

Key management:

Clara Raposo, President
João Manso Neto, CEO
Miguel Valente, CFO

Ownership:

Altri (c.17%), Promendo
Investimentos (c.10%), Actium
Capital (c.10%), Caderno Azul
(c.9%), Liverfluxo (c.8%), V-
Ridium Europe, S.P. Z0.0.
(c.8%), 1Thing (c.5%), Free
Float (c.37%)

Figure 1: GreenVolt Corporate
Governance

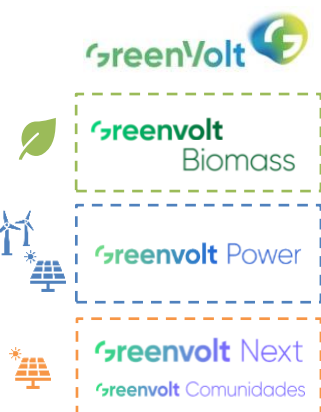


Figure 2: GreenVolt Business
Segments

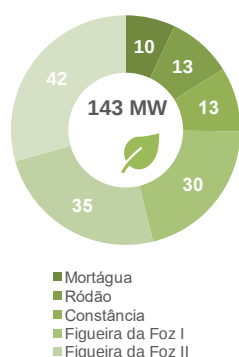


Figure 3: GreenVolt Biomass Installed
Capacity (MW). Source: GreenVolt

Company overview

GreenVolt – Energias Renováveis (or “the Company”) is a **developer** and **independent power producer** (“IPP”), focused on **regulated biomass** and **expanding its presence into solar PV and wind** through promising pan-European pipeline. In 2021, the Company was rebranded as GreenVolt and executed an **IPO**, at a **share price of €4.25** and **market cap over €500m**.

Business segments

GreenVolt’s strategy is based on three business areas, **biomass**, **solar PV** and **onshore wind**, and **distributed energy generation** (“DG”), which operate under four brands, in light of the November 2022 image rebranding (Figure 2).

The Company is currently the **leading biomass player in Portugal**, where it operates **five biomass plants**, which use 100% forestry residues, with a total installed capacity of **101MW**. It benefits from a **Feed-in-Tariff (“FIT”)**¹ regime and **long-term supply agreements** with Altri, one of the leading eucalyptus pulp producers in Europe, which contribute to a higher competitive advantage, when compared to other biomass players. In 2019, GreenVolt made its first **scaling-up move in Europe**, by acquiring a 51% stake in the Tilbury biomass plant in the **United Kingdom (“UK”)**, representing **42MW** of installed capacity, being solely sourced by urban waste wood. The six plants culminate in a **total installed capacity of 143MW**, which produce, on an annual basis, over **1,000GWh** (Figure 3). As of 2021, this business segment contributed with **c.€131m** of revenues (representing **93%** of total revenues).

GreenVolt is committed to grow well beyond biomass through other renewable energy sources (“RES”), as **solar PV and wind**. In fact, the acquisition of **V-Ridium Power Group (“V-Ridium”)** in 2021 has set an ambitious pace for the Company. In Q3 2022, the Company had a **portfolio of c.6.7GW**, of which 229MW are in operation or UC, 56MW RtB, c.3.1GW in advance stage, and c.3.3GW early stage (Figure 4).² As of 2021, the segment contributed with **1%** of total revenues. The Romanian solar park acquired in the second quarter of 2022 (LIONS Solar PV plant), for **€83m**, set the grounds for the **first operational solar PV and wind capacity** of GreenVolt, amounting to **45MWp**.

Together with the development of these large-scale projects, GreenVolt is dedicated to DG, focused on **self-consumption** and **energy communities**. By September 2022, GreenVolt had **33.5MWp** of installed capacity and **112.3MWp** of backlog. It contributed with **c.€8m** of revenues (**6%** of total revenues) in 2021.

Geographic footprint

Regarding the **biomass segment** (Figure 5), GreenVolt is expected to continue to leverage **Portugal** and **UK’s** plants, having been identified potential acquisitions of two power plants in the UK (c.40MW) in the next five years. In Portugal, not only is a new Mortágua plant with 10MW of installed capacity expected to become operational in the next years, but the Company is considering greenfield opportunities up to 20MW as well.³

GreenVolt is present in **10 countries** in **Europe** through its **solar PV and wind business** (Figure

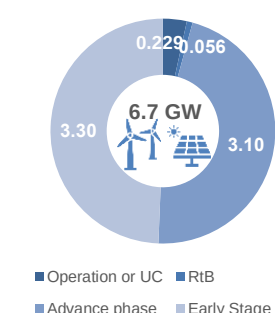


Figure 4: GreenVolt Solar PV and
Wind Installed Capacity and existing
pipeline (GW). Source: GreenVolt

¹ Feed-in-Tariff is a scheme that aims to support the development of renewable energy sources by setting a guaranteed, above market price for producers

² The pipeline status is divided in three stages: (i.) Under Construction (“UC”), (ii.) Ready to Build (“RtB”), (iii.) advanced stage, and (iv.) early stage. As the name suggests, under construction means the construction already started or it is about to start, while the advance and early stages correspond to a period before licencing

³ The new Mortágua plant is under technical evaluation (several authorisations have already been issued)

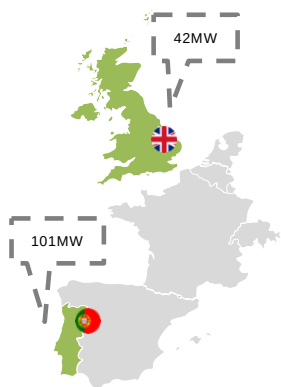


Figure 5: GreenVolt Biomass Installed Capacity as of Q3 2022. Source: GreenVolt

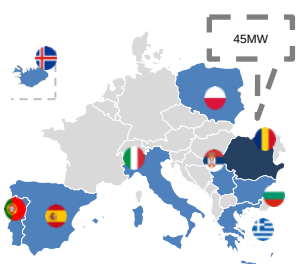


Figure 6: GreenVolt Solar PV and Wind Installed Capacity (dark blue) and existing pipeline (light blue) as of Q3 2022. Source: GreenVolt



Figure 7: GreenVolt Distributed Energy Generation Installed Capacity as of Q3 2022. Source: GreenVolt

6). **Poland** and **Portugal** are the two countries where the Company has the largest pipeline (c.3.4GW and c.1GW, respectively) and the only ones with projects UC (112MW and 72MW, respectively). GreenVolt seeks to identify opportunities in project-scarce markets with high growth potential geographies, by identifying market niches. Recently entered in the **United States of America**, with the acquisition of an 80% stake in Oak Creek, a California-based renewable energy projected developer, through V-Ridium. The Company is also present in Germany, through the minority stake in MaxSolar, acquired in 2022.

Concerning **DG** (Figure 7), the Company is bound to focus on **Portugal**, **Spain**, and Greece, especially the first two, where self-consumption penetration remains significantly below other European countries.

Value chain

The **biomass value chain** commences with harvesting resources, intermediate storage, energy production, and energy distribution.⁴ GreenVolt positions itself in the **production of energy** part, but the overall competitive advantage also arises due to favourable conditions relating to the other steps of the value chain. For example, GreenVolt has a supply agreement with Altri, which secures the totality of resources used in the Portuguese biomass plants, and all biomass plants are established in proximity to its raw material suppliers, which minimises transport and storage costs (Figure 8).⁵

The **renewable energy value chain** begins with the development of projects, ranging from their design, execution and construction, to the operation of the respective plants (Figure 9). Since the profitability and the level of investment required are inversely proportional to the stage of maturity of the project, GreenVolt has preferred to operate mainly in the **most upstream part of the value chain, (project development)**, and only **opportunisticly** in the construction and operations stages. Project development, which is less capital intensive, but highly challenging in terms of know-how and expertise, includes all activities from the design of a project to its authorisation.

Strategy plan (2022e-2026e)

A year after the IPO, and post Altri spin-off, GreenVolt **updated its original strategy**, shifting its strategic positioning from biomass to solar PV and wind and DG, in light of growth potential, although the former is currently still the main driver in terms of cashflow generation.^{6,7}

GreenVolt is focused to **continually optimise current biomass plants**, only investing in new ones in an opportunistic manner, implying a **lower investment than expected until 2026e**, when compared to the Company's goals at IPO.

Concerning solar PV and wind, on one hand, GreenVolt seeks to **capture sky-rocketing electricity prices** and secure them into the future through **Power Purchasing Agreements⁸ ("PPAs")**, and on the other, it seeks to focus on the development and opportunistic construction of renewable assets, both translating into an **expected higher investment until 2026e**, in comparison to what was defined originally. Regarding the former, GreenVolt aims to **fully develop some projects to be held in its balance sheet (c.20% to 30% of existing pipeline)** (Figure 10). Regarding the latter, and as mentioned previously, GreenVolt pursues the value



Figure 8: GreenVolt - Biomass Value Chain

⁴ <https://www.intechopen.com/chapters/58145>

⁵ Potential conflicts of interest could arise because Altri is both a relevant shareholder (c.17%) and supplier

⁶ On May 25th, 2022, Altri, at the time GreenVolt's main shareholder, distributed a dividend in kind of the 43% stake it held directly, maintaining a qualified stake

⁷ https://greenvolt.pt/wp-content/uploads/2022/06/2022.06.09_Greenvolt_Deal-roadshow-presentation_Final-1.pdf

⁸ Power Purchase Agreement is an electricity supply contract between an energy generator and a counterparty



Figure 9: GreenVolt – Solar PV and Wind Value Chain

chain position where the highest return lies. Therefore, it will position **most of its pipeline (c.70% to 80% of existing pipeline) in the project development phase, meaning more projects will be driven to a Commercial Operations Date (“COD”) or RtB phases** (50:50 split between the two). Selling at COD means selling the projects once they have been built, while RtB means selling the projects with all permits/authorizations to start the construction (Figure 11). **Projects are analysed on a case-to-case basis**, and the decision to go COD or remain RtB is based on **value maximization**. However, GreenVolt is expecting a **higher demand for COD projects**, as a result of shortage of raw materials, volatility of commodity prices and supply chain bottlenecks, which increases CapEx. Although driving projects to COD is riskier, GreenVolt’s track record puts the Company well positioned to do so successfully.

In the DG segment, GreenVolt also intends to allocate **higher than expected investment until 2026e**, as a response to the accelerated growth on the self-consumption and energy communities markets, especially in Iberia.

This updated business plan, which entails a larger pipeline and expected overall investment, also means an **updated and ambitious CapEx plan until 2026e**, covering all verticals: it grew from c.€1.5bn-€1.8bn to **c.€3.8bn-€4.2bn** (Figure 12). Additionally, for this revised strategy, the recent **€100m capital increase** and **€150m green bonds issuance** also play a great role, even though additional debt will be required, as we will discuss in the Discount Rate section.⁹

In terms of value, GreenVolt expects to present a **c.43% 2021a-2026e EBITDA CAGR**, being **EBITDA mainly driven by solar PV and wind (55%)**, followed by biomass (30%) and DG (15%), by 2026e.

Macroeconomic context

Russia’s war on Ukraine, which started earlier this year, has pivoted a **global energy crisis**, due to **geopolitical volatility** and **constrained access to natural gas**, leading to **energy prices spikes**. This, in turn, has stressed the need for **energy resilience** (in terms of supply and independence), and is acting as a push for RES.¹⁰ Moreover, current market instability, namely high levels of inflation, has **increased the demand for long-term PPAs while shifting investor’s appetite for projects at a COD stage**, which are fully de-risked, as mentioned previously. Even though this trend seems favourable to GreenVolt’s strategy plan, we do not discard possible shock waves related to supply chain bottleneck which could downgrade sales in the asset rotation strategy. As a result, the future seems volatile and hard to predict.

Industry overview

Global energy demand has been traditionally linked to **economic growth** and **energy prices**.¹¹ In fact, under the Net Zero Scenario (“NZS”), global energy demand will serve an economy more than twice as large and a population with 2bn more people.¹² Nonetheless, contrary to popular expectations, **global energy demand in 2050 is c.8% smaller than today** driven by (i.) **the entrance of new technologies** (which lead to a pronounced increase in energy efficiency and

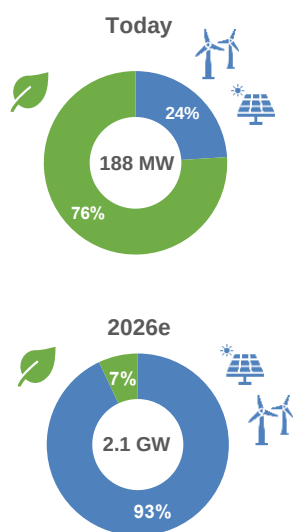


Figure 11: Operational Capacity Mix by Technology. Source: GreenVolt



Figure 10: Asset Rotation Strategy

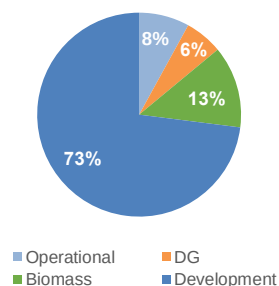


Figure 12: Investments per Technology 2026e. Source: GreenVolt

⁹ <https://greenvolt.com/wp-content/uploads/2022/11/FR84179.pdf>

¹⁰ <https://www.mckinsey.com/~/media/McKinsey/Industries/Oil%20and%20Gas/Our%20Insights/Global%20Energy%20Perspective%202022/Global-Energy-Perspective-2022-Executive-Summary.pdf>

¹¹ https://link.springer.com/chapter/10.1007/978-3-030-86884-0_26

¹² https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroBy2050-ARoadmapfortheGlobalEnergySector_CORR.pdf

rise of electrification), (ii.) **policies**, and (iii.) a **shift in consumer preferences**.^{12,13,14}

Regarding energy prices, studies have shown, as expected, that the price-elasticity of energy demand follows the behaviour of microeconomics: **higher prices lead to lower consumption (hence lower demand)**, and vice-versa.¹⁵ Though, the **day-ahead wholesale price determination** introduces a peculiar case for renewables. For the determination system in a **non-discriminatory auction** (i.e., marginal bid), producers submit supply schedules for each hour of the day-ahead and power plants are ranked by lowest to highest bid, forming a **generation stack**. Then, in a typical generation stack, the baseload plants (i.e., the first dispatched) tend to be renewables (e.g., hydro, solar, wind), as its **marginal costs are virtually zero**. Only then do mid-merit plants and peaking plants play a role (e.g., natural gas, coal), being the last dispatched producer the one which sets the price, and all winning bidders get paid the same clearing price. Thus, even if global energy demand fluctuates, renewables are expected to be always consumed (i.e., demand for renewable energy should not be as impacted). On the other hand (i.e., the supply side), it is worthwhile to mention the **supply concerns that arise from rising levels of generation from technologies with virtually zero marginal costs**, such as renewables, which will unavoidably reduce prices, such that **revenues will not cover investments**, and therefore **discourage energy supply from RES**. Notwithstanding, we believe the actual pricing system can be improved, as research as demonstrated (through market or government incentives), so the supply of energy generated from RES should not be as impacted.¹⁶

Although global energy demand may be reaching a **plateau level**, the three drivers highlighted above point to a **shift in the energy mix towards a greater contribution of renewables**, hence driving **renewable** energy demand.

In fact, globally, **demand for fossil fuels is expected to decline considerably** and the **global fuel mix is expected to shift substantially** between 2020 and 2050. Coal usage, which reached its peak in 2014, is bound to decrease by c.15%. Having dropped steeply in 2020 due to the pandemic, oil demand recovered swiftly, returning to a pre-pandemic level and is expected to reach a plateau shortly after 2030. Natural gas demand is expected to increase from c.4bn cubic metres (cm) in 2020 to c.5.7bn cm in 2050. Nuclear energy will expand by 15% between 2020 and 2030, mainly reflecting expansions in China.¹⁷ Moreover, **the energy sector is expected to be based mainly on renewable energy**: more than half of total energy supply in 2050 stems from wind, solar, bioenergy, geothermal and hydro energy.¹⁷

This **positive RES demand outlook has a favourable impact on GreenVolt's revenues**, as we will further analyse in the industry revenue drivers section.

Industry revenue drivers: analysis and forecasts

Based on McKinsey's revenue forecast top-down approach, we will **analyse industry-specific and overall economic trends** in the next sections, based on historical financial analysis but mostly based on how GreenVolt is positioned for the future (i.e., in light of the updated strategy plan), and we will **forecast the drivers we believe to have the most (quantifiable) impact**.

To do so, we will split the analysis in a threefold manner, by assessing to which extent: **(i.) does**

¹³ <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-decoupling-of-gdp-and-energy-growth-a-ceo-guide>

¹⁴ <https://corporate.exxonmobil.com/energy-and-innovation/outlook-for-energy/energy-demand>

¹⁵ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7375269/>

¹⁶ [https://www.iit.comillas.edu/documentacion/IIT-21-008A_preprint/Zero-marginal-](https://www.iit.comillas.edu/documentacion/IIT-21-008A_preprint/Zero-marginal-cost_electricity_market_designs_lessons_learned_from_hydro_systems_in_Latin_America_might_be_applicable_for_decarbonization.pdf)

[cost_electricity_market_designs_lessons_learned_from_hydro_systems_in_Latin_America_might_be_applicable_for_decarbonization.pdf](https://www.iit.comillas.edu/documentacion/IIT-21-008A_preprint/Zero-marginal-cost_electricity_market_designs_lessons_learned_from_hydro_systems_in_Latin_America_might_be_applicable_for_decarbonization.pdf)

¹⁷ https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroBy2050-ARoadmapfortheGlobalEnergySector_CORR.pdf

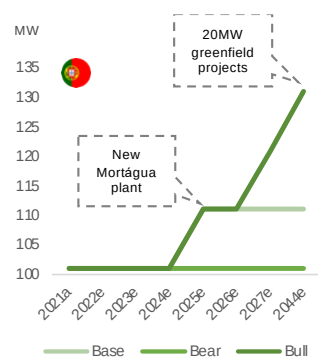


Figure 13: GreenVolt's installed capacity of biomass plants in Portugal – actual and forecasted. Source: Analysts' estimates

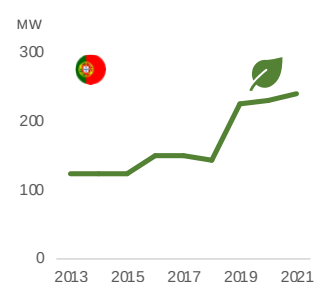


Figure 14: Cumulative Installed Capacity Biomass in Portugal. Source: DGE

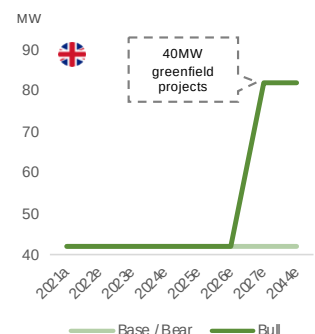


Figure 15: GreenVolt's installed capacity of biomass plants in UK – actual and forecasted. Source: Analysts' estimates

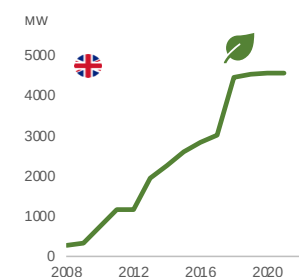


Figure 16: Cumulative Installed Capacity Biomass in United Kingdom. Source: Statista

GreenVolt have the required portfolio to capture market share (i.e., installed capacity and load factor); **(ii.) do other competitors have the required portfolio to capture market share;** and **(iii.) is GreenVolt affected by long-term trends** (i.e., average selling price, regulatory frameworks, and other macroeconomics factors, as exchange rates, inflation, and real GDP growth).¹⁸ As expected, **prices** and **quantities** will play a decisive role, both in terms of revenues and costs, as we will discuss later on. This analysis will be conducted on a **scenario basis**: **(i.) base**, where we assume an economy state as is, **(ii.) bear**, where we anticipate a negative outlook from GreenVolt's standpoint, and **(iii.) bull**, where we anticipate a more optimistic prospect.

■ Installed capacity

Biomass

In Portugal, in the **base** scenario, we include **10MW from the new Mortágua power plant concession under development**, which expects the production license to be granted during 2022, and once the production is granted, it will take approximately 2 years for the new Mortágua Power Plant to be operational (i.e., in **2025e**). In this case, the Company starts with 101MW of installed capacity, jumps to 111MW in 2025e, and the installed capacity remains unaltered going forward.

In the **bear** scenario, we **disregard the new Mortágua plant** (hence, installed capacity equals to 101MW each forecasted year). In the **bull** scenario, not only we include the new Mortágua plant in 2025e, but we **include additional 20MW of greenfield projects** foreseen in GreenVolt's strategy (10MW in 2027e and 10MW in 2029e, following the rationale of 2-years needed for a plant to become operational). In this optimistic outlook, installed capacity amounts to 101MW from 2022e-2024e, 111MW from 2025e-2026e, 121MW from 2027e-2028e, and 131MW from 2029e onwards (Figure 13).

In this country, there is a total of 240MW installed capacity of biomass without cogeneration (Figure 14).¹⁹ There are five main competitors and GreenVolt is the **current leader, with a 42% of market share**, as of October 2022. The runner-up is The Navigator Company, and the remaining market is fragmented, thus offering the opportunity to further consolidate. On one hand, the Portuguese National Energy and Climate Plan 2021-2030 ("**PNEC2030**") **sets a very shy evolution of installed capacity for biomass plants**: a growth of 100MW from 2020 to 2030.²⁰ On the other, since the **market is highly dispersed**, we do not believe the extra 100MW of expected installed capacity will be a major drawback to GreenVolt's market share in the base and bull scenarios, in Portugal.

As for the UK, in the **base** and **bear** scenarios, we expect **installed capacity to remain unaltered** (42MW each year). In the **bull** scenario, we factor in additional 40MW of installed capacity due to **greenfield projects in 2027e** (given the 2022e-2026e guidance from the Company mentioned earlier) (Figure 15).

In this country, total installed capacity of operational biomass plants amounts to c.4.5GW, as of September 2021, which implies a **mere 1% of market share for the Tilbury plant**.²¹ On one hand, from 2018 to 2021, **total installed capacity has remained somewhat steady**, having only

¹⁸ Valuation – measuring and managing the value of companies, McKinsey, 6th edition

¹⁹ <https://www.dgeg.gov.pt/media/bn3ky3fc/dgeg-arr-2022-10.pdf>

²⁰ <https://bcdportugal.org/wp-content/uploads/2020/12/PNEC-2030-Plano-Nacional-Energia-e-Clima.pdf>

²¹ <https://www.statista.com/statistics/792421/cumulative-plant-biomass-capacity-united-kingdom/>

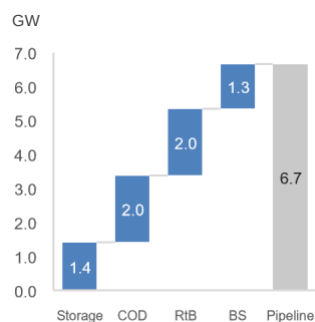


Figure 17: Pipeline breakdown by 2026e. Source: Analysts' estimates

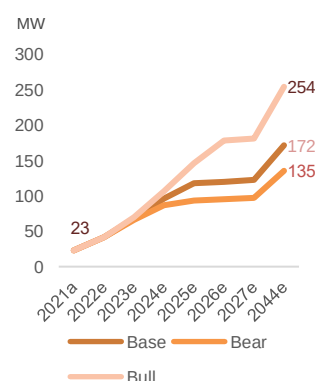


Figure 18: GreenVolt's installed capacity of distributed generation – actual and forecasted. Source: Analysts' estimates

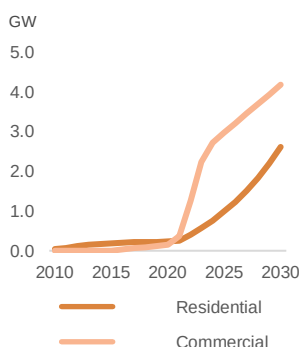


Figure 19: Cumulative PV Solar Installed Capacity in Portugal 2010-2030. Source: Bloomberg

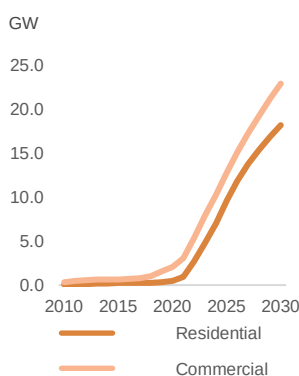


Figure 20: Cumulative PV Solar Installed Capacity in Spain 2010-2030. Source: Bloomberg

increase by 0.2% from 2020 to 2021 (Figure 16). On the other hand, **biomass plants are typically geographically dispersed across the country**, and there are no other plants close to Tilbury. Thus, we do not believe in any major threat to GreenVolt's market share in the UK, in any scenario under analysis.

Solar PV and wind

This segment's portfolio has 45MW of installed capacity, which pales in comparison with the c.6.7GW of pipeline. Thus, and since we are not factoring in further acquisitions to current installed capacity, we will only forecast the current **pipeline's trajectory into installed capacity per se** (i.e., fully operational).

By **2026e**, we forecast all current pipeline will become **RtB or COD** (within the asset rotation strategy) or **operational** (within the balance sheet strategy), based on the current Company's data on the split between technology, country, and pipeline stage.²² From c.6.7GW of existing pipeline, we forecast 75% will be under the asset rotation strategy, while the remainder will be held on the balance sheet.

Regarding the asset rotation scheme, by **2026e**, we estimate **c.5.4GW** will be either at RtB or COD stages, including the **c.1.4GW** from storage. As for the assets to be held on the balance sheet, we forecast the Company will keep assets in countries on which it has more **expertise** (i.e., Portugal, Poland, and Greece), and keep the already operational solar PV plant in Romania. Thus, by 2023e, we expect c.60MW to be held on balance sheet to be fully operational (i.e., generating cash inflows), and by **2026e**, we expect this number to reach **c.1.3GW**, thus growing by c.22x (Figure 17).

Since the majority of the existing pipeline is early stage, we are not able to draw meaningful conclusions regarding GreenVolt market share in this segment.

Distributed energy generation

As of September 2022, the Company had 33.5MWp of installed capacity and we expect to reach **c.41.9MWp** in the end of 2022, across all scenarios, which represents a growth of 82% from 2021a to 2022e.

From 2023e-2026e, we assume GreenVolt will be able to sustain a high growth rate (though lower than 82%), since (i.) it is the **ramp-up phase of this segment**, and (ii.) the **Iberian countries are GreenVolt's key targets** (Figure 18).

Regarding the latter driver, in Portugal, the low-voltage distribution grid has seen an **expanding connection of small-scale distributed generation and generation intended for self-consumption (both mainly stemming from solar PV)**. From 2015 to 2021, **total installed capacity of small-scale distributed generation grew** from c.18MW to c.61MW (Figure 19). Moreover, in Spain, total installed capacity of both residential and commercial reached c.4.0GW in 2021 (Figure 20). Nonetheless, while sunny Iberian countries seem like it would be innate front-runners in Europe for rooftop solar PV, the region has lingered when compared to other European nations, such as Germany and Italy, which had c.61GW and c.23GW in 2021, respectively.²³ Therefore, **the Iberia presents great potential**, since it has a considerably high horizontal irradiation resource, **yet little current penetration**, which will be favourable for a new-entrant

²² <https://greenvolt.com/wp-content/uploads/2022/11/fsd1239109.pdf>

²³ Bloomberg

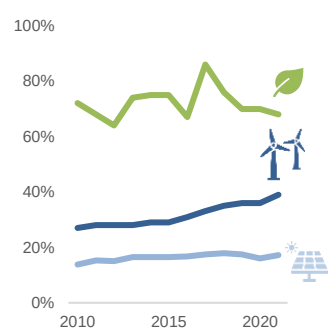


Figure 21: Load Factor by Technology. Source: IRENA

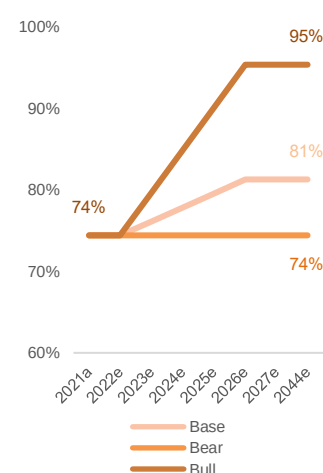


Figure 22: GreenVolt's load factor of biomass plants in Portugal – actual and forecasted. Source: Analysts' estimates

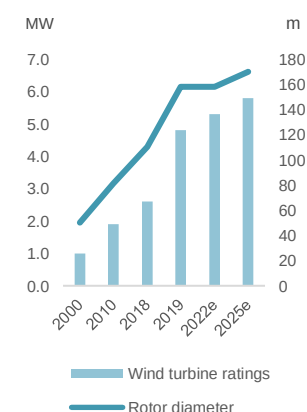


Figure 23: Global average hub height and respective rotor diameter per year. Key characteristics to improve energy yield. Source: IRENA

player like GreenVolt, implying a **positive outlook for its market share** in the short term.

After the explicit forecast period, from 2027e onwards, we believe **current government policies and incentives will drive more players to join the self-consumption and energy communities market**. Therefore, the aggregate market is expected to **mature** and follow economic growth, and as such, installed capacity is assumed to grow at the European Union real GDP growth level forecasted for each year.

■ Load factor

Load factor, also known as capacity factor, offers a glimpse on the reliability of power plants: it measures how often a plant is running at maximum power and it is given by **energy exported over maximum production possible, as per license**.²⁴ This key parameter is crucial to measure plants' functioning, and it **depends mainly on the technology used, the availability of resources, and shutdown periods for maintenance**. According to IRENA, comparing the three technologies, **biomass has the highest load factor**, followed by onshore wind and solar PV. Notwithstanding, these last two followed an upward trend due to technology advancements, while biomass followed a trendless period peaking in 2017 with a capacity factor of 87% (Figure 21).²⁵

Biomass

Load factors for biomass plants are usually very high but come at the expense of regular maintenance work (as boilers often face technical problems), and are highly reliant on continuous access to biomass supply. GreenVolt's power plants in Portugal are powered completely by forestry residues through a long-term agreement by Altri Madeira, and in the UK by urban waste wood through a long-term contract with Esken Renewables. **A clear advantage that contributed to an overall average load factor improvement over the past years.**

In Portugal, in the **base** scenario, we assume that **load factor increases by 7pp until 2026e, to c.81%**, since this amount would have been the **average Portuguese load factor as of 2021, if we excluded the Ródão outage**, whose plant load factor would be 71% instead of 40%. In a **bull** scenario, we assume **load factor increases by 22pp until 2026e, to c.96%**, since we consider GreenVolt will be able to apply its **Figueira da Foz II know-how** (i.e., use of maximum quality materials, boiler adjustments, among others) in the remaining Portuguese biomass plants, and therefore we expect all load factors to converge to Figueira da Foz II current load factor. In these two scenarios, from 2027e-2044e, load factor remains stable (i.e., growth of 0%). In the **bear** scenario, the no-growth assumption (i.e., no improvement) begins at 2022e (Figure 22).

As for the **UK** load factor (i.e., Tilbury plant), it is already **above the national average load factor** (89% vs. 68%, respectively), therefore **we do not expect improvement (nor deterioration) under in any scenario**.²⁶

In any case, **current load factors (Portugal and UK) are well above the overall worldwide average in 2021 (68%)**, which suggests a favourable outlook for GreenVolt's ability to maintain its competitive advantage going forward and, as a result, its market share.

Solar PV and wind / distributed energy generation

Globally, **onshore wind load factor increased 44% from 2010 to 2021**. It depends essentially

²⁴ <https://www.energy.gov/ne/articles/what-generation-capacity#:~:text=The%20Capacity%20Factor&text=It%20basically%20measures%20how%20often,of%20the%20time%20in%202021.>

²⁵ IRENA - Renewable Power Generation – Costs in 2021

²⁶ <https://www.statista.com/statistics/555734/plant-biomass-electricity-load-factor-uk/>

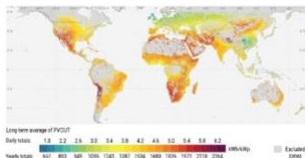


Figure 24: Solar Photovoltaic Power Potential by Country. Source: World Bank

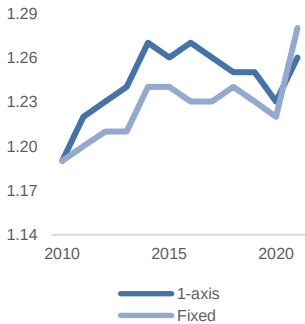


Figure 25: Global average inverter load ratio trend, 2010-2021. Source: IRENA

on two factors: quality and speed of the air, and turbine technology used. Locations that provide incoming wind consistently, such as mountainous areas, larger and more reliable turbines, along with higher hub-heights and larger rotor diameters, increase load factors (Figure 23).²⁵

The global weighted average **solar PV capacity factor rose from 14% in 2010 to 17% in 2021**. Solar load factor depends essentially on solar resource conditions and technology used. The upward load factor trend is related to (i.) technology advancements, such as the adoption of bifacial technology – i.e., power is produced from both sides of the panel when illuminated, which increases electricity generation – and the use of trackers – which allow the panel to follow the sun's movements and consequently generates more solar power; (ii.) construction of solar PV in sunnier locations, quantified by the global horizontal irradiation (GHI), which changes significantly per country (Figure 24); and (iii.) inverter load ratio (ILR), also known as the DC-to-AC ratio, inverters running close to full capacity lead to a higher load factor (Figure 25).

Overall, we believe **technology will continue to improve**, consequently rising load factors in these two renewable technologies. Although we lack information on current GreenVolt's load factor of these two technologies, we believe it is well positioned to also **surf the technological improvements wave**, thus maintaining competitive advantage going forward (and market share, as a consequence).

Competitive landscape

In light of the Company's **business mix, geographic exposure**, and, more importantly, the **stage of pipeline**, we believe it is **difficult to find a true competitor** within the **market of energy generation from RES**. Nonetheless, GreenVolt may potentially face competition from **several fronts** which can compromise its ability to keep or increase market share.

On one hand, we have **traditional oil & gas players** (e.g., Repsol, Galp, BP) and **integrated utilities** (e.g., Iberdrola, Endesa, EDP, Engie), and on the other, we have **renewable players focused on solar PV and wind** (e.g., EDP Renováveis, Acciona, Orsted) and **traditional biomass players** (e.g., Albioma and Drax). Since GreenVolt positioning lies in between traditional biomass players and pure renewable players, it emerges with **unique** characteristics, such as, **expertise on biomass** (market leader in Portugal), **asset rotation** (due to V-Ridium), and on **developing plants** to RtB or COD.

As a result, we defined two peer groups, **Peer Group 1** (Figure 26) and **Peer Group 2** (Figure 27), to better understand if **competitors have the required portfolio to capture market share**. We considered firms with market cap between €0.8bn-€2bn and present in Europe (an exception to this is EDP Renováveis, with market cap above €20bn, because we deemed interesting to include a Portuguese peer). Notwithstanding, we note key differences. For example, regarding Albioma, half of its biomass installed capacity is overseas, and Ence and Drax are not pure players (the former has a pulp business and the latter has other generation sources) (Table 1). As for solar PV and wind players, we highlight the discrepancy of installed capacity between GreenVolt and the other peers: GreenVolt has 45MW (solar PV), whereas Audax, the peer with lowest market cap, has 180MW (solar PV and wind) (Table 2).

Biomass

In Portugal, forestry biomass is mainly linked to paper industry. Like GreenVolt, The Navigator Company, the biggest paper firm in Portugal, also takes advantage of pulp facilities for feedstock. In December 2020, it **inaugurated a biomass power station in Figueira da Foz next to its**



Figure 26: Peer Group 1 - companies active in the biomass generation - Albioma, Drax and Ence



Figure 27: Peer Group 2 - companies mostly focused on solar PV and wind development - Audax, EDP Renováveis, and Voltalia

	Albioma	Drax	Ence
Country			
Market Cap (€m)	1,526	2,165	855
Total IC (MW)	1,029	3,200	266
Share IC Europe	58%	100%	100%

Table 1: Key figures - Peer Group 1.
Source: Companies' websites

	Audax	EDPR	Voltaia
Country			
Market Cap (€m)	335	20,479	1,979
Total IC (MW)	180	13,590	1,129
Share IC Europe	89%	42%	23%

Table 2: Key figures - Peer Group 2.
Source: Companies' websites

Power Plant	Price (€/MWh)
Constância	118.3
Figueira da Foz I	120.3
Figueira da Foz II	116.3
Mortágua	132.0
Ródão	122.5

Table 3: GreenVolt FiT Portugal 2021

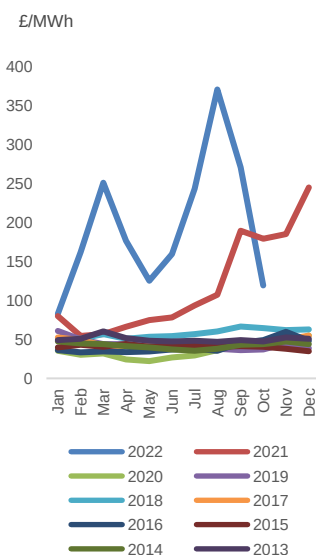


Figure 28: N2ex Day Ahead Auction Prices in the UK 2013-2022. Source: Nordpool

pulp mill. Nevertheless, given the low growth forecasts of installed capacity for biomass in Portugal, the high barriers to entry associated with high capital costs and the fact that the Navigator Company is not showing signs of future market growth in the biomass industry, we do not believe other players will be able to displace GreenVolt's market share in Portugal.

On the other hand, given the **recent Albioma acquisition by KKR**, which may hint further investments in the company, and **Drax's £3bn investment plan** in biomass endeavours, we believe further expansions in Europe would be needed to outdo competition (peers' average installed capacity amounts to c.1.3GW, whereas GreenVolt's amounts to 143MW).^{27,28} Though, in light of the current strategy plan, GreenVolt does not foresee any major expansions, which may help these competitors to capture market share. This, in turn, will be reflected in the overall growth rate for the segment (or lack thereof), which we will further discuss in the RONIC section.

Solar PV and wind / distributed energy generation

There have been several technology improvements in the solar industry which created new products, such as **solar shingles**, **solar windows** and **floating solar panels**. In terms of **innovation** and **use of break-through technologies**, EDP and EDP Renováveis have been front-runners. Regarding the latter, on earlier this year, EDP launched a **floating solar PV power plant** in Alqueva with c.12k photovoltaic panels and an annual production capacity of **c.7.5GWh**.²⁹ Moreover, EDP has a venture capital firm, **EDP Ventures**, specialized in investing in early-stage tech start-ups working in energy, being one of its investment priorities the distributed energy resources. **Offshore wind** has been the latest trend in wind technology, although GreenVolt relies only on onshore, currently. EDP Renováveis through **Ocean Winds**, a joint venture with Engie, is focused on offshore wind farms, with 1.5GW in operation and 16.6GW of total gross capacity.³⁰ On the other hand, for example, **Voltaia** presents an investment plan wherein it allocates c.85% of c.€2.5bn-€3.0bn (which pales in comparison when compared to GreenVolt's CapEx plan until 2026e, as mentioned previously).³¹ Thus, not all competitors will be able to capture market share, but there are major players, as EDP Renováveis, that could pose a serious threat.

- Average selling price

Biomass

GreenVolt's main revenue driver stems from the sale of electricity produced at the plants to the grid, through **PPAs** with guaranteed remuneration: the **FiT** regime in Portugal and the **Renewable Obligation Certificates (ROCs)** in the UK.³²

In Portugal, in the **base** and **bull** scenarios, **average selling price (€/MWh) from 2022e-2026e corresponds to the average FiT during 2021 (€121.9/MWh)** (Table 3); from **2027e-2044e, we expect the average selling price to converge to the national average indicative tariff (€119.0/MWh)**.^{33,34}

²⁷ https://media.kkr.com/rss-feed/news-release/?news_id=257162f3-2f33-4968-acae-5680c956238a&type=1

²⁸ <https://www.drax.com/wp-content/uploads/2022/03/Drax-2021-FYR-Analyst-Presentation-website.pdf>

²⁹ <https://www.edp.com/en/innovation/floating-solar-pannels>

³⁰ <https://www.oceanwinds.com/ocean-winds-in-numbers/>

³¹ <https://www.voltaia.com/static-files/c3f5feed-9ce7-4f6c-8d9b-e6b5a32e0f34>

³² Additionally, in the UK, there is also a variable component that changes according to the evolution of the electricity price (Brown Power)

³³ <https://www.dgeg.gov.pt/pt/areas-setoriais/energia/energia-eletrica/producao-de-energia-eletrica/producao-potencia-instalada-superior-a-1-mva/remuneracao-das-energias-renovaveis/>

³⁴ For a coefficient Z=9.6

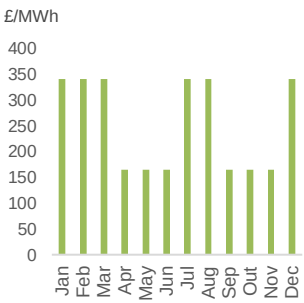


Figure 29: Monthly selling prices for 2023 and 2024 for the UK. Source: Analysts' estimates

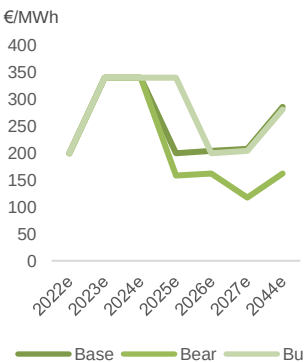


Figure 30: GreenVolt's average prices of biomass in the UK – forecasted. Source: Analysts' estimates

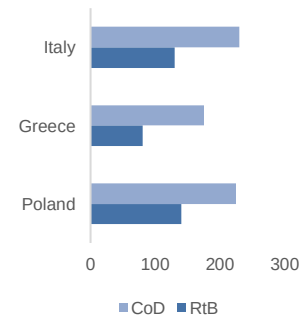


Figure 31: Solar - Project Exit Value per MW per country. Source: GreenVolt

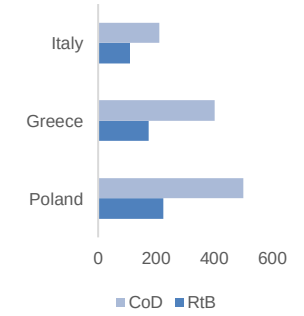


Figure 32: Wind - Project Exit Value per MW per country. Source: GreenVolt

In both these scenarios, we assume FiT regimes applicable to each plant are extended, in light of the recent 17-year extension for the Mortágua plant. For the bear scenario, we do not expect FiT extensions and, as such, GreenVolt will only be entitled to the fixed remuneration until FiT expiry; upon termination, the Company becomes subject to market prices. The first FiT expiry occurs in 2031e, so in 2032e we considered an average selling price of €60.5/MWh, which corresponds to the EEX Spanish Power Futures, provided by the European Energy Exchange, with a predetermined future date of 2032.³⁵ We assumed this futures contract would be a good proxy for Portugal, since the Spanish market is close to the Portuguese one (MIBEL). Notwithstanding, we understand this is a non-arbitrage price and that futures contracts are mainly used to hedge risk but, as the Federal Reserve Bank of St. Louis pointed out, they may have a side advantage of often suggesting good price forecasts.³⁶ From that year onwards, the average selling price will grow at the forecasted inflation level.

In the UK, according to the Office of Gas and Electricity Markets (Ofgem), the Renewables Obligation (RO) is a support mechanism for large-scale renewable electricity projects, and attributes an obligation on UK electricity suppliers to provide an increasing proportion of electricity they supply from renewable sources; failure to meet this obligation implies a buy-out payment.³⁷ Since there is no historical data on GreenVolt's ability (or lack thereof) to meet its obligations, we assume a no-failure case. Thus, the market price is the only relevant driver for the average selling price (Figure 28). For 2022e, we assume the average N2ex Day Ahead Auction Prices in the UK, from January until November 2022 (£199.7/MWh) under all scenarios.³⁸

Given the current unprecedented and volatile times, the UK government set an Energy Price Guarantee (effective as of October 2022) with a cap of £340/MWh until April 2024e.³⁹ It is challenging to forecast energy prices during normal times, but it is even more so during an energy crisis. Chris O'Shea, from Centrica, owner of British Gas, said that it is expected that for the next two years, prices will continue at an all-time high.⁴⁰ As a result, for all scenarios, in 2023e and 2024e, we considered that the cap will be met on the summer and winter months and, for the remaining, we used the average spring and autumn months of 2022a (Figure 29). Only in the bull scenario, we considered that the cap will be extended until 2025e.

From 2027e onwards, we considered a return to pre-Ukraine war electricity price levels, since current forecasts and research do not suggest otherwise. For the base and bull scenarios, we assumed the same selling price as for 2022e, which will be indexed to the forecasted inflation for each year going forward. As for the more severe scenario, we believe selling prices will return to N2ex Day Ahead Auction Prices in the UK for 2021 (£117.4/MWh) (Figure 30).

Solar PV and wind

Revenue streams from this business segment are threefold: (i.) provision of accounting, administrative and asset management services, (ii.) sale of solar and wind energy projects, in a RtB or COD phase, and (iii.) fixed remuneration schemes from the plants developed to own. We consider the first stream to be non-meaningful, when compared to the other two.

³⁵ <https://www.eex.com/en/market-data/power/futures/#%7B%22snippetpicker%22%3A%2234%22%7D> – as of November 4, 2022
³⁶ <https://www.stlouisfed.org/publications/regional-economist/january-2002/the-futures-market-as-forecasting-tool-an-imperfect-crystal-ball#:~:text=Although%20futures%20contracts%20primarily%20exist,also%20produce%20good%20price%20forecasts.>
³⁷ <https://www.ofgem.gov.uk/environmental-and-social-schemes/renewables-obligation-ro>
³⁸ <https://www.nordpoolgroup.com/en/Market-data/1/GB/Auction-prices/UK/monthly/?view=table>
³⁹ <https://www.ofgem.gov.uk/information-consumers/energy-advice-households/check-if-energy-price-cap-affects-you>
⁴⁰ <https://bionic.co.uk/blog/noticed-energy-prices-have-fallen-over-past-few-months/>

Country	Solar	Wind
Portugal ¹	38.0	n.a.
Poland ²	71.4	48.3
Greece ²	54.0	63.0
Romania ³	37.0	n.a.

Table 4: Remunerations per country (€/MWh). Note: ¹ – PPA; ² – CfD; ³ – Market Price. Source: Roedl, Balkan Green Energies, GreenVolt Capital Increase Prospectus

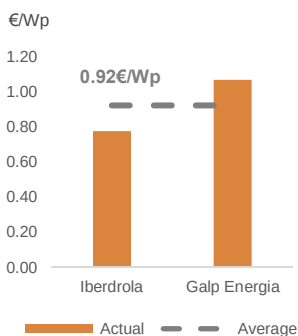


Figure 33: Distributed Generation - B2B - Selling Price (Peer analysis). Source: Companies' website

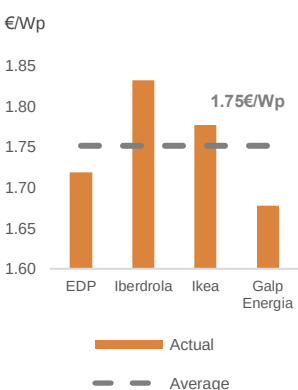


Figure 34: Distributed Generation - B2C - Selling Price (Peer analysis). Source: Companies' website

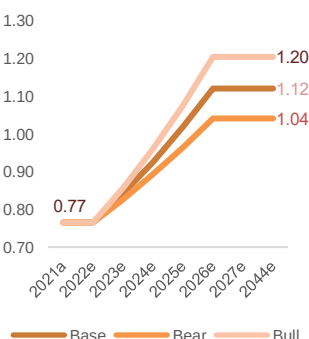


Figure 35: GreenVolt's average prices of distributed generation – actual and forecasted. Source: Analysts' estimates

Regarding the asset rotation strategy, GreenVolt recently executed its **first sale solar PV and wind agreement** with Iberdrola, for **€155m**, of a portfolio of assets (2 wind farms and 6 solar PV farms) under construction in Poland, with an installed capacity of 98MW.^{41,42} Since this is the first sale, we do not have much visibility to extrapolate forecasts. However, the Company shared **value creation schemes** (detailing development costs and exit values), for **both stages** (RtB and COD) and for **each technology** (solar PV and wind), for Poland, Greece, and Italy.⁴³ Hence, we extrapolated from this available information **averages of selling prices (i.e., the exit values)** for each technology and each stage, as seen in Figure 31 and Figure 32.

On the other hand, given its deep knowledge and industry expertise, GreenVolt wants to maintain a portfolio of contracted assets in its balance sheet. These projects have remunerations based on **PPA or Contract-for-Difference (CfD)**, which mitigates the risk and hedge against energy market price fluctuations.⁴⁴ To estimate the average selling price, we checked in-place remuneration schemes for each country and for each technology, which are summarized in Table 4.

Distributed energy generation

As for distributed generation, GreenVolt's revenues mainly come from **the installation and maintenance of distributed solar energy production units (B2B and B2C)**, and also from the development and financing of projects to improve energy efficiency through solar energy.

Regarding the former, due to the lack of information concerning prices set by the Company, **we extrapolated the average selling price from GreenVolt's revenues and installed capacity as of September 2022: €0.77/Wp or c.€765k/MWp.**⁴⁵

To assess the reasonability of such extrapolation, we conducted a **peer analysis in the Portuguese market, in both market segments**. For the **B2B segment**, we considered prices from Iberdrola and Galp Energia, resulting in an average of **€0.92/Wp** (Figure 33). For the **B2C segment**, we considered comparable offerings from EDP, Iberdrola, IKEA, and Galp Energia, resulting in an average of **€1.75/Wp** (Figure 34). Since we lack visibility regarding the weight of each segment in GreenVolt's offerings, we applied an equal weighted average, resulting in **€1.34/Wp or €1.340k/MWp**. We concluded the extrapolation is fair, given GreenVolt is a new entrant in the installation of solar panels market (i.e., setting low prices to penetrate the market), and **we set the average selling price for 2022e as c.€765k/MWp**.

In all cases, we assume GreenVolt will be able to set higher prices in the short-term (2023e-2026e), as it gains traction, **approaching the overall market**, at different growth rates, but never surpassing it. Selling prices will grow by 10pp, 8pp, and 12pp each year in the short-term, in the base, bear and bull scenarios, respectively. Thus, average selling prices by 2026e are expected to be €1.120k/MWp, €1.040k/MWp, and €1.204k/MWp, in the base, bear, and bull scenarios, respectively. From 2027e onwards, prices will sustain in all scenarios (Figure 35).

Regulatory framework

⁴¹ All the assets in the portfolio are covered by the PPA signed with T-Mobile Polska

⁴² <https://econews.pt/2022/08/01/greenvolt-offloads-solar-wind-farms-in-poland-to-iberdrola-for-e155m/>

⁴³ https://greenvolt.com/wp-content/uploads/2022/06/pdf_en_20.pdf

⁴⁴ CfD is based on a difference between the market price and an agreed strike price. If the latter is higher than the former, the CfD counterparty must pay the renewable producer the difference (IEA, 2019)

⁴⁵ <https://greenvolt.pt/wp-content/uploads/2022/09/Greenvolt2Q22PT.pdf>

Main regulatory framework impacting GreenVolt
Paris Climate Agreement
European Green Deal
Clean Energy Package
"Plano Nacional Integrado Energia Clima 2030"
Spanish National Integrated Energy and Climate Plan 2021-2030
European Law on Climate
FIT for 55
Climate Framework Law
REPowerEU Plan

Table 5: Energy regulations impacting GreenVolt

The **Paris Climate Agreement**, an international treaty on climate change, whose chief goal is to limit global warming, for which the transition towards renewables such as solar PV and wind play a critical role, has already yielded positive outcomes. After more than 5 years after the Agreement, **c.70% of the countries involved disclosed quantified renewable energy targets**, which can be read as a driver to foster energy demand for these technologies.^{46,47} Additionally, since the introduction of the **EU Renewable Energy Directive** (2009/28/EC) in 2009, **the use of renewables has been increasing yearly**, reaching more than 22% in 2020, which puts the EU in a leading position for renewables technology improvement and deployment.⁴⁸ The European Commission has recently presented the **REPowerEU Plan**, its reaction to the adversities and global energy market disruption caused by Russia's war on Ukraine. Measures foreseen in the plan include **the broadening of energy supplies, and quicker roll-out of renewable energy to replace fossil fuels** in homes, industry, and power generation.⁴⁹ Many other agreements and laws have been passed and enacted towards a **future where RES play a far greater role than today** (Table 5).

Regarding **incentive schemes**, we acknowledge that the transition towards climate neutrality requires a wider policy context that integrates a broad range of instruments, such as **remuneration mechanisms** and **subsidies**. Concerning the latter, energy subsidies in the EU reached an amount to c.€180bn in 2019. Despite EU's pledges to phase out fossil fuel subsidies, research shows they are not decreasing adequately. Nonetheless, **renewable energy subsidies amounted to c.€80bn, an improvement of 8% since 2015**.⁵⁰ Regarding the former, mechanisms as FIT and ROC play a key role in incentivizing energy generation through RES, since energy markets alone cannot bring the targeted level of renewables in the EU, requiring **national governments to enact support schemes to foster increased investment in renewable energy**.⁵¹

Overall, between agreements, laws and incentive schemes, we believe the current state of affairs to be very bright in what RES are concerned, which impacts GreenVolt positively.

- Other macroeconomic factors: exchange rates, inflation and real GDP growth

GreenVolt has **pan-European operations** and is consequently subject to **exchange rate fluctuations**, so **foreign-currency risk** must be incorporated in the valuation.⁵² In fact, in the event of a currency exchange rate adversely changing, it may potentially lead to **(economic) value destruction**. In particular, we highlight the operations in the **UK**, through the **Tilbury plant**, and in **Poland**, through **V-Ridium**, which represent a high percentage of EBITDA and pipeline, respectively. Therefore, to convert cashflow projections we will use the **forward-rate method**, as suggested by McKinsey: we will forecast foreign cash flows in the foreign currency, and convert these into euros using the relevant forward exchange rates. We expect the **Pound Sterling (GBP) forward exchange rate to EUR** and **Polish Zloty (PLN) forward exchange rate to EUR**, for **2022e**, to be **1.14** and **0.21**, respectively, based on current forwards, as per data from the

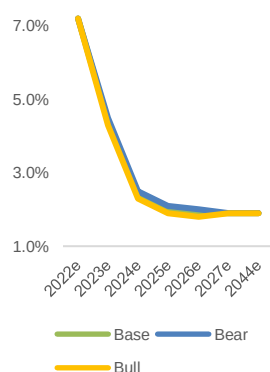


Figure 36: Inflation forecast. Source: IMF

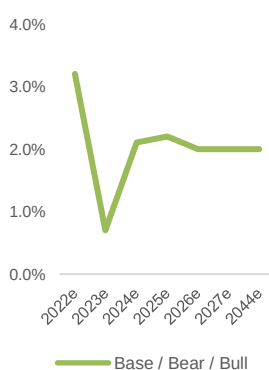


Figure 37: EU real GDP growth forecast. Source: IMF

⁴⁶ <https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement>

⁴⁷ <https://www.irena.org/publications/2020/Dec/Renewable-energy-and-climate-pledges>

⁴⁸ https://energy.ec.europa.eu/topics/renewable-energy/renewable-energy-directive-targets-and-rules/renewable-energy-directive_en#:~:text=The%20directive%20establishes%20a%20new,the%2020%25%20target%20for%202020.

⁴⁹ https://ec.europa.eu/commission/presscorner/detail/en/IP_22_3131

⁵⁰ https://www.ffms.pt/sites/default/files/2022-07/state_of_the_energy_union_report_2021.pdf

⁵¹ https://energy.ec.europa.eu/topics/renewable-energy/financing/support-schemes_en

⁵² Valuation – measuring and managing the value of companies, McKinsey, 6th edition

Financial Times.^{53,54} Since we lack visibility for the next years, 2022e will be the best forecasted year, and from 2023e-20244e, we assume the same value.

Based on IMF predictions for advanced economies, **we forecast inflation to be at 7.2% in 2022e in all scenarios.**⁵⁵ For **2023e-2026e**, in the **base** case we continued to assume IMF predictions; our forecasts are **1pp above and 1pp below IMF's** in the **bear** and **bull** scenarios, respectively. **From 2027e onwards, we assume 1.9% for each forecasted year, in all scenarios.** In this later stage (i.e., 2027e-2044e), we believe the Company will be **more mature** and thus **able to increase selling prices at the inflation level**, enough to overcome higher capital costs as well as operating costs (Figure 36).

For the EU real GDP growth, and since we only consider this factor to be relevant in a more mature stage (i.e., 2027e-2044e), when we believe **GreenVolt will be able to grow quantities** (e.g., DG installed capacity) **at the economy level**, we forecasted this growth to be equal to the one forecasted by IMF for **2026e (2%)** (Figure 37).⁵⁶

⁵³ <https://markets.ft.com/data/currencies/tearsheet/summary?s=GBPEUR>

⁵⁴ <https://markets.ft.com/data/currencies/tearsheet/summary?s=PLNEUR>

⁵⁵ <https://www.imf.org/external/datamapper/PCPIPCH@WEO/ADVEC>

⁵⁶ https://www.imf.org/external/datamapper/NGDP_RPCH@WEO/EU

GREENVOLT, S.A.

ENERGY RENEWABLES

STUDENT: BÁRBARA JANEIRO & FRANCISCA PAULOS

COMPANY REPORT

16 DECEMBER 2022

49107@novasbe.pt / 29120@novasbe.pt

GREENVOLT: THE PROMISING PIPELINE

Generating power... and value

- GreenVolt's equity story is based on three pillars: (i.) a successful **biomass scaling-up move in the UK** after building a **strong base in Portugal**, (ii.) **promising solar PV and wind pipeline**, as a result of V-Ridium's acquisition, and (iii.) **ramping up** on the decentralized energy generation opportunity.
- While biomass has been the **main revenue contributor historically**, we expect a **modest growth going forward**, and a ROIC slightly above WACC, as the Company **shifts its strategy towards solar PV and wind energy generation assets**.
- We view the strategic shift as **positive**, as it will allow GreenVolt to **consolidate its position as a renewable European powerhouse**, which will be the chief value driver in stock returns in the mid-term. **Potential catalysts** to our rating include FiT extensions in Portuguese biomass assets and ability to capture sky-rocketing wholesale electricity prices in the UK, while **potential risks** include a low successful rate in the asset rotation / develop to own strategy and management dependency.
- We initiate coverage at **Buy**, issuing a Target Price of **€9.50**, implying an **upside of c.17%**. We derive value using a **SOTP approach**: biomass and distributed energy generation via **DCF** (WACC of 5.71% / 5.24% and perpetuity growth rate of 0.57% / 2.00%, respectively) and solar PV and wind segment via **EV/MW** (WACC of 5.24%).

Company description

GreenVolt is a Portuguese-based energy renewable company which develops and operates a portfolio of energy generation plants and assets under three segments: biomass, solar PV and wind, and distributed energy generation. It has a pan-European portfolio of pipeline (c.6.7GW) and installed capacity (c.188MW).

Recommendation: **BUY**

Total Shareholder Return 16.90%

Price Target FY23: **9.50 €**

Price (as of 14-Dec-22): **8.13 €**

Bloomberg: GVOLT PL, Reuters: GVOLT.LS

52-week range (€) 5.25-10.80

Market Cap (€m) 1,176.10

Outstanding Shares (m) 139.17

Source: Bloomberg



Source: Bloomberg; Note: rebased at 100

(Values in € millions)	2021	2022E	2023E
Revenues	140.6	257.6	327.9
EBITDA	57.0	106.2	173.7
Net Profit	8.1	22.8	63.4
EPS	0.10	0.22	0.47

Source: Bloomberg

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Founded: 2002

Headquarters: Porto, Portugal

Key management:

Clara Raposo, President

João Manso Neto, CEO

Miguel Valente, CFO

Ownership:

Altri (c.17%), Promendo Investimentos (c.10%), Actium Capital (c.10%), Caderno Azul (c.9%), Liverfluxo (c.8%), V-Ridium Europe, S.P. Z.O.O. (c.8%), 1Thing (c.5%), Free Float (c.37%)

Figure 1: GreenVolt Corporate Governance

Company overview

GreenVolt – Energias Renováveis (or “the Company”) is a **developer** and **independent power producer** (“IPP”), focused on **regulated biomass** and **expanding its presence into solar PV and wind** through promising pan-European pipeline. In 2021, the Company was rebranded as GreenVolt and executed an **IPO**, at a **share price of €4.25** and **market cap over €500m**.

Business segments

GreenVolt's strategy is based on three business areas, **biomass**, **solar PV** and **onshore wind**, and **distributed energy generation** (“DG”), which operate under four brands, in light of the November 2022 image rebranding (Figure 2).

The Company is currently the **leading biomass player in Portugal**, where it operates **five biomass plants**, which use 100% forestry residues, with a total installed capacity of **101MW**. It benefits from a **Feed-in-Tariff (“FIT”)**¹ regime and **long-term supply agreements** with Altri, one of the leading eucalyptus pulp producers in Europe, which contribute to a higher competitive advantage, when compared to other biomass players. In 2019, GreenVolt made its first **scaling-up move in Europe**, by acquiring a 51% stake in the Tilbury biomass plant in the **United Kingdom (“UK”)**, representing **42MW** of installed capacity, being solely sourced by urban waste wood. The six plants culminate in a **total installed capacity of 143MW**, which produce, on an annual basis, over **1,000GWh** (Figure 3). As of 2021, this business segment contributed with **c.€131m** of revenues (representing **93%** of total revenues).

GreenVolt is committed to grow well beyond biomass through other renewable energy sources (“RES”), as **solar PV and wind**. In fact, the acquisition of **V-Ridium Power Group (“V-Ridium”)** in 2021 has set an ambitious pace for the Company. In Q3 2022, the Company had a **portfolio of c.6.7GW**, of which 229MW are in operation or UC, 56MW RtB, c.3.1GW in advance stage, and c.3.3GW early stage (Figure 4).² As of 2021, the segment contributed with **1%** of total revenues. The Romanian solar park acquired in the second quarter of 2022 (LIONS Solar PV plant), for **€83m**, set the grounds for the **first operational solar PV and wind capacity** of GreenVolt, amounting to **45MWp**.

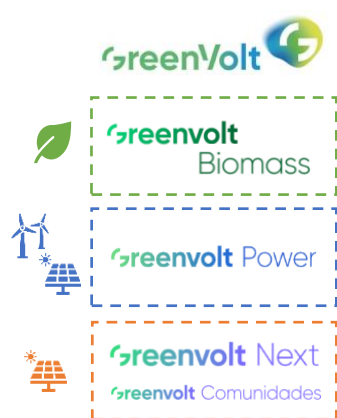


Figure 2: GreenVolt Business Segments

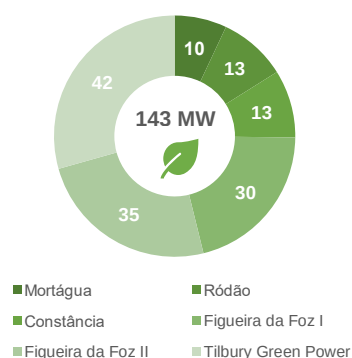


Figure 3: GreenVolt Biomass Installed Capacity (MW). Source: GreenVolt

¹ Feed-in-Tariff is a scheme that aims to support the development of renewable energy sources by setting a guaranteed, above market price for producers

² The pipeline status is divided in three stages: (i.) Under Construction (“UC”), (ii.) Ready to Build (“RtB”), (iii.) advanced stage, and (iv.) early stage. As the name suggests, under construction means the construction already started or it is about to start, while the advance and early stages correspond to a period before licencing

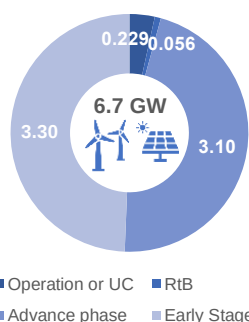


Figure 4: GreenVolt Solar PV and Wind Installed Capacity and existing pipeline (GW). Source: GreenVolt

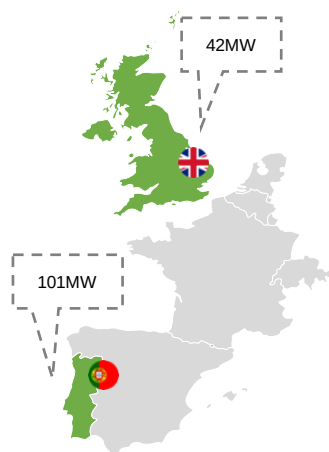


Figure 5: GreenVolt Biomass Installed Capacity as of Q3 2022. Source: GreenVolt

Together with the development of these large-scale projects, GreenVolt is dedicated to DG, focused on **self-consumption** and **energy communities**. By September 2022, GreenVolt had **33.5MWp** of installed capacity and **112.3MWp** of backlog. It contributed with **c.€8m** of revenues (**6%** of total revenues) in 2021.

Geographic footprint

Regarding the **biomass segment** (Figure 5), GreenVolt is expected to continue to leverage **Portugal** and **UK's** plants, having been identified potential acquisitions of two power plants in the UK (c.40MW) in the next five years. In Portugal, not only is a new Mortágua plant with 10MW of installed capacity expected to become operational in the next years, but the Company is considering greenfield opportunities up to 20MW as well.³

GreenVolt is present in **10 countries** in **Europe** through its **solar PV and wind business** (Figure 6). **Poland** and **Portugal** are the two countries where the Company has the largest pipeline (c.3.4GW and c.1GW, respectively) and the only ones with projects UC (112MW and 72MW, respectively). GreenVolt seeks to identify opportunities in project-scarce markets with high growth potential geographies, by identifying market niches. Recently entered in the **United States of America**, with the acquisition of an 80% stake in Oak Creek, a California-based renewable energy projected developer, through V-Ridium. The Company is also present in Germany, through the minority stake in MaxSolar, acquired in 2022.

Concerning **DG** (Figure 7), the Company is bound to focus on **Portugal, Spain**, and **Greece**, especially the first two, where self-consumption penetration remains significantly below other European countries.

Value chain

The **biomass value chain** commences with harvesting resources, intermediate storage, energy production, and energy distribution.⁴ GreenVolt positions itself in the **production of energy** part, but the overall competitive advantage also arises due to favourable conditions relating to the other steps of the value chain. For example, GreenVolt has a supply agreement with Altri, which secures the totality of resources used in the Portuguese biomass plants, and all biomass plants are established in proximity to its raw material suppliers, which minimises transport and storage costs (Figure 8).⁵

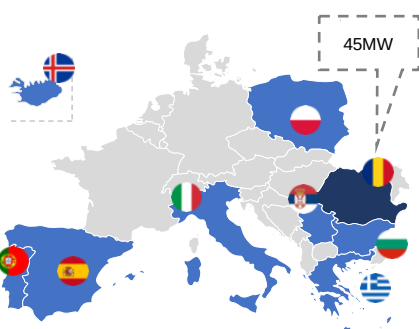


Figure 6: GreenVolt Solar PV and Wind Installed Capacity (dark blue) and existing pipeline (light blue) as of Q3 2022. Source: GreenVolt

³ The new Mortágua plant is under technical evaluation (several authorisations have already been issued)

⁴ <https://www.intechopen.com/chapters/58145>

⁵ Potential conflicts of interest could arise because Altri is both a relevant shareholder (c.17%) and supplier

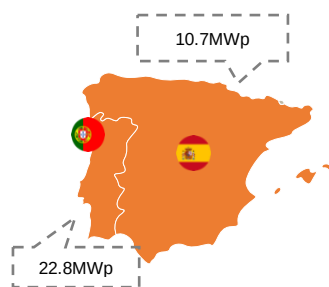


Figure 7: GreenVolt Distributed Energy Generation Installed Capacity as of Q3 2022. Source: GreenVolt

The **renewable energy value chain** begins with the development of projects, ranging from their design, execution and construction, to the operation of the respective plants (Figure 9). Since the profitability and the level of investment required are inversely proportional to the stage of maturity of the project, GreenVolt has preferred to operate mainly in the **most upstream part of the value chain, (project development)**, and only **opportunistically** in the construction and operations stages. Project development, which is less capital intensive, but highly challenging in terms of know-how and expertise, includes all activities from the design of a project to its authorisation.

Strategy plan (2022e-2026e)

A year after the IPO, and post Altri spin-off, GreenVolt **updated its original strategy**, shifting its strategic positioning from biomass to solar PV and wind and DG, in light of growth potential, although the former is currently still the main driver in terms of cashflow generation.^{6,7}

GreenVolt is focused to **continually optimise current biomass plants**, only investing in new ones in an opportunistic manner, implying a **lower investment than expected until 2026e**, when compared to the Company's goals at IPO.

Concerning solar PV and wind, on one hand, GreenVolt seeks to **capture sky-rocketing electricity prices** and secure them into the future through **Power Purchasing Agreements⁸ ("PPAs")**, and on the other, it seeks to focus on the development and opportunistic construction of renewable assets, both translating into an **expected higher investment until 2026e**, in comparison to what was defined originally. Regarding the former, GreenVolt aims to **fully develop some projects to be held in its balance sheet (c.20% to 30% of existing pipeline)** (Figure 10). Regarding the latter, and as mentioned previously, GreenVolt pursues the value chain position where the highest return lies. Therefore, it will position **most of its pipeline (c.70% to 80% of existing pipeline) in the project development phase, meaning more projects will be driven to a Commercial Operations Date ("COD") or RtB phases** (50:50 split between the two). Selling at COD means selling the projects once they have been built, while RtB means selling the projects with all permits/authorizations to start the construction (Figure 11). **Projects are analysed on a case-to-case basis**, and the decision to go COD or remain RtB is based on **value maximization**. However, GreenVolt is expecting a **higher demand for COD projects**, as a result of shortage of raw materials,



Figure 8: GreenVolt - Biomass Value Chain

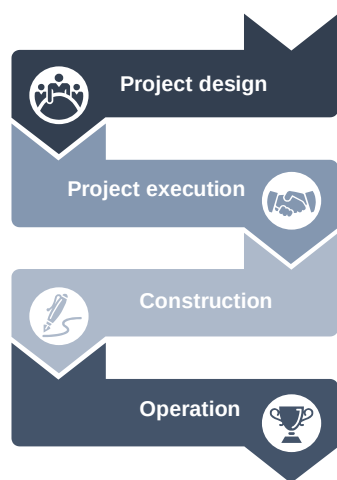


Figure 9: GreenVolt – Solar PV and Wind Value Chain

⁶ On May 25th, 2022, Altri, at the time GreenVolt's main shareholder, distributed a dividend in kind of the 43% stake it held directly, maintaining a qualified stake

⁷ https://greenvolt.pt/wp-content/uploads/2022/06/2022.06.09_Greenvolt_Deal-roadshow-presentation_Final-1.pdf

⁸ Power Purchase Agreement is an electricity supply contract between an energy generator and a counterparty

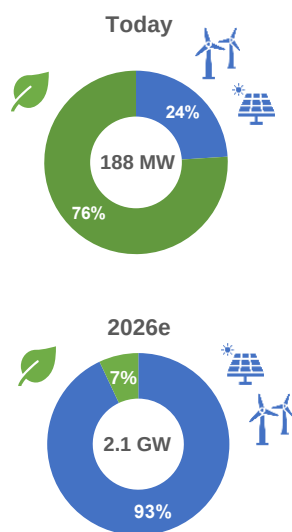


Figure 10: Operational Capacity Mix by Technology. Source: GreenVolt

volatility of commodity prices and supply chain bottlenecks, which increases CapEx. Although driving projects to COD is riskier, GreenVolt's track record puts the Company well positioned to do so successfully.

In the DG segment, GreenVolt also intends to allocate **higher than expected investment until 2026e**, as a response to the accelerated growth on the self-consumption and energy communities markets, especially in Iberia.

This updated business plan, which entails a larger pipeline and expected overall investment, also means an **updated and ambitious CapEx plan until 2026e**, covering all verticals: it grew from c.€1.5bn-€1.8bn to **c.€3.8bn-€4.2bn** (Figure 12). Additionally, for this revised strategy, the recent **€100m capital increase** and **€150m green bonds issuance** also play a great role, even though additional debt will be required, as we will discuss in the Discount Rate section.⁹

In terms of value, GreenVolt expects to present a **c.43% 2021a-2026e EBITDA CAGR**, being **EBITDA mainly driven by solar PV and wind (55%)**, followed by biomass (30%) and DG (15%), by 2026e.

Macroeconomic context



Figure 11: Asset Rotation Strategy

Russia's war on Ukraine, which started earlier this year, has pivoted a **global energy crisis**, due to **geopolitical volatility** and **constrained access to natural gas**, leading to **energy prices spikes**. This, in turn, has stressed the need for **energy resilience** (in terms of supply and independence), and is acting as a push for RES.¹⁰ Moreover, current market instability, namely high levels of inflation, has **increased the demand for long-term PPAs while shifting investor's appetite for projects at a COD stage**, which are fully de-risked, as mentioned previously. Even though this trend seems favourable to GreenVolt's strategy plan, we do not discard possible shock waves related to supply chain bottleneck which could downgrade sales in the asset rotation strategy. As a result, the future seems volatile and hard to predict.

Industry overview

Global energy demand has been traditionally linked to **economic growth** and **energy prices**.¹¹ In fact, under the Net Zero Scenario ("NZS"), global energy demand will serve an economy more than twice as large and a population with 2bn

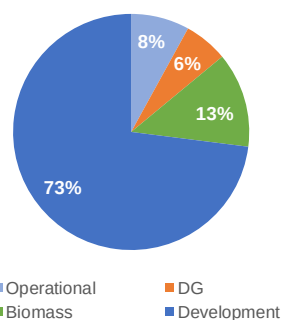


Figure 12: Investments per Technology 2026e. Source: GreenVolt

⁹ <https://greenvolt.com/wp-content/uploads/2022/11/FR84179.pdf>

¹⁰ <https://www.mckinsey.com/~/media/McKinsey/Industries/Oil%20and%20Gas/Our%20Insights/Global%20Energy%20Perspective%202022/Global-Energy-Perspective-2022-Executive-Summary.pdf>

¹¹ https://link.springer.com/chapter/10.1007/978-3-030-86884-0_26

more people.¹² Nonetheless, contrary to popular expectations, **global energy demand in 2050 is c.8% smaller than today** driven by (i.) **the entrance of new technologies** (which lead to a pronounced increase in energy efficiency and rise of electrification), (ii.) **policies**, and (iii.) **a shift in consumer preferences**.^{12,13,14}

Regarding energy prices, studies have shown, as expected, that the price-elasticity of energy demand follows the behaviour of microeconomics: **higher prices lead to lower consumption (hence lower demand)**, and vice-versa.¹⁵ Though, the **day-ahead wholesale price determination** introduces a peculiar case for renewables. For the determination system in a **non-discriminatory auction** (i.e., marginal bid), producers submit supply schedules for each hour of the day-ahead and power plants are ranked by lowest to highest bid, forming a **generation stack**. Then, in a typical generation stack, the baseload plants (i.e., the first dispatched) tend to be renewables (e.g., hydro, solar, wind), as its **marginal costs are virtually zero**. Only then do mid-merit plants and peaking plants play a role (e.g., natural gas, coal), being the last dispatched producer the one which sets the price, and all winning bidders get paid the same clearing price. Thus, even if global energy demand fluctuates, renewables are expected to be always consumed (i.e., demand for renewable energy should not be as impacted). On the other hand (i.e., the supply side), it is worthwhile to mention the **supply concerns that arise from rising levels of generation from technologies with virtually zero marginal costs**, such as renewables, which will unavoidably reduce prices, such that **revenues will not cover investments**, and therefore **discourage energy supply from RES**. Notwithstanding, we believe the actual pricing system can be improved, as research as demonstrated (through market or government incentives), so the supply of energy generated from RES should not be as impacted.¹⁶

Although global energy demand may be reaching a **plateau level**, the three drivers highlighted above point to a **shift in the energy mix towards a greater contribution of renewables**, hence driving **renewable** energy demand.

In fact, globally, **demand for fossil fuels is expected to decline considerably** and the **global fuel mix is expected to shift substantially** between 2020 and 2050. Coal usage, which reached its peak in 2014, is bound to decrease by c.15%. Having dropped steeply in 2020 due to the pandemic, oil demand recovered swiftly, returning to a pre-pandemic level and is expected to reach a plateau shortly after

¹² https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroBy2050-ARoadmapfortheGlobalEnergySector_CORR.pdf

¹³ <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-decoupling-of-gdp-and-energy-growth-a-ceo-guide>

¹⁴ <https://corporate.exxonmobil.com/energy-and-innovation/outlook-for-energy/energy-demand>

¹⁵ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7375269/>

¹⁶ https://www.iit.comillas.edu/documentacion/IIT-21-008A_preprint/Zero-marginal-cost_electricity_market_designs_lessons_learned_from_hydro_systems_in_Latin_America_might_be_applicable_for_decarbonization.pdf

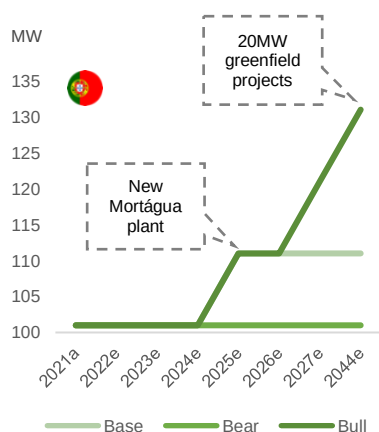


Figure 13: GreenVolt's installed capacity of biomass plants in Portugal – actual and forecasted. Source: Analysts' estimates

2030. Natural gas demand is expected to increase from c.4bn cubic metres (cm) in 2020 to c.5.7bn cm in 2050. Nuclear energy will expand by 15% between 2020 and 2030, mainly reflecting expansions in China.¹⁷ Moreover, **the energy sector is expected to be based mainly on renewable energy**: more than half of total energy supply in 2050 stems from wind, solar, bioenergy, geothermal and hydro energy.¹⁷

This **positive RES demand outlook has a favourable impact on GreenVolt's revenues**, as we will further analyse in the industry revenue drivers section.

Industry revenue drivers: analysis and forecasts

Based on McKinsey's revenue forecast top-down approach, we will **analyse industry-specific and overall economic trends** in the next sections, based on historical financial analysis but mostly based on how GreenVolt is positioned for the future (i.e., in light of the updated strategy plan), and we will **forecast the drivers we believe to have the most (quantifiable) impact**.

To do so, we will split the analysis in a threefold manner, by assessing to which extent: **(i.) does GreenVolt have the required portfolio to capture market share** (i.e., installed capacity and load factor); **(ii.) do other competitors have the required portfolio to capture market share**; and **(iii.) is GreenVolt affected by long-term trends** (i.e., average selling price, regulatory frameworks, and other macroeconomics factors, as exchange rates, inflation, and real GDP growth).¹⁸ As expected, **prices and quantities** will play a decisive role, both in terms of revenues and costs, as we will discuss later on. This analysis will be conducted on a **scenario basis**: **(i.) base**, where we assume an economy state as is, **(ii.) bear**, where we anticipate a negative outlook from GreenVolt's standpoint, and **(iii.) bull**, where we anticipate a more optimistic prospect.

■ Installed capacity

Biomass

In Portugal, in the **base** scenario, **we include 10MW from the new Mortágua power plant concession under development**, which expects the production license to be granted during 2022, and once the production is granted, it will take approximately 2 years for the new Mortágua Power Plant to be operational (i.e., in **2025e**). In this case, the Company starts with 101MW of installed capacity, jumps to 111MW in 2025e, and the installed capacity remains unaltered going forward.

¹⁷ https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroBy2050-ARoadmapfortheGlobalEnergySector_CORR.pdf

¹⁸ Valuation – measuring and managing the value of companies, McKinsey, 6th edition

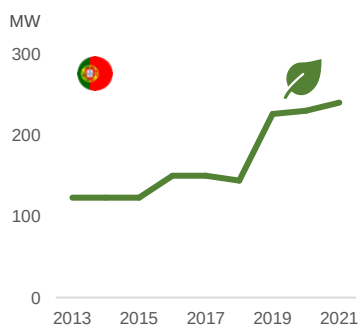


Figure 14: Cumulative Installed Capacity Biomass in Portugal. Source: DGE

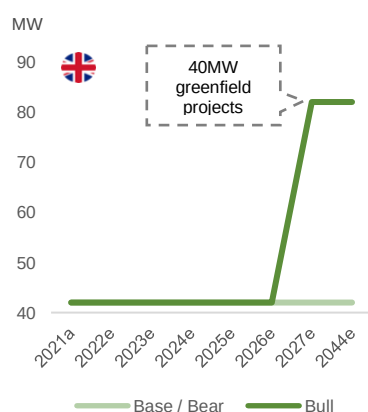


Figure 15: GreenVolt's installed capacity of biomass plants in UK – actual and forecasted. Source: Analysts' estimates

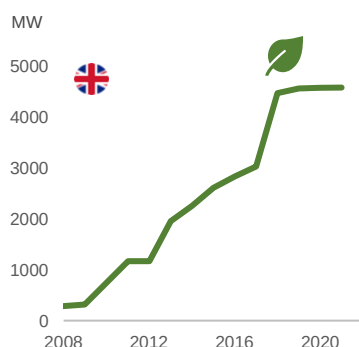


Figure 16: Cumulative Installed Capacity Biomass in United Kingdom. Source: Statista

In the **bear** scenario, we **disregard the new Mortágua plant** (hence, installed capacity equals to 101MW each forecasted year). In the **bull** scenario, not only we include the new Mortágua plant in 2025e, but we **include additional 20MW of greenfield projects** foreseen in GreenVolt's strategy (10MW in 2027e and 10MW in 2029e, following the rationale of 2-years needed for a plant to become operational). In this optimistic outlook, installed capacity amounts to 101MW from 2022e-2024e, 111MW from 2025e-2026e, 121MW from 2027e-2028e, and 131MW from 2029e onwards (Figure 13).

In this country, there is a total of 240MW installed capacity of biomass without cogeneration (Figure 14).¹⁹ There are five main competitors and GreenVolt is the **current leader, with a 42% of market share**, as of October 2022. The runner-up is The Navigator Company, and the remaining market is fragmented, thus offering the opportunity to further consolidate. On one hand, the Portuguese National Energy and Climate Plan 2021-2030 ("**PNEC2030**") **sets a very shy evolution of installed capacity for biomass plants**: a growth of 100MW from 2020 to 2030.²⁰ On the other, since the **market is highly dispersed**, we do not believe the extra 100MW of expected installed capacity will be a major drawback to GreenVolt's market share in the base and bull scenarios, in Portugal.

As for the UK, in the **base** and **bear** scenarios, we expect **installed capacity to remain unaltered** (42MW each year). In the **bull** scenario, we factor in additional 40MW of installed capacity due to **greenfield projects in 2027e** (given the 2022e-2026e guidance from the Company mentioned earlier) (Figure 15).

In this country, total installed capacity of operational biomass plants amounts to c.4.5GW, as of September 2021, which implies a **mere 1% of market share for the Tilbury plant**.²¹ On one hand, from 2018 to 2021, **total installed capacity has remained somewhat steady**, having only increase by 0.2% from 2020 to 2021 (Figure 16). On the other hand, **biomass plants are typically geographically dispersed across the country**, and there are no other plants close to Tilbury. Thus, we do not believe in any major threat to GreenVolt's market share in the UK, in any scenario under analysis.

Solar PV and wind

This segment's portfolio has 45MW of installed capacity, which pales in comparison with the c.6.7GW of pipeline. Thus, and since we are not factoring in

¹⁹ <https://www.dgeg.gov.pt/media/bn3ky3fc/dgeg-arr-2022-10.pdf>

²⁰ <https://bcdportugal.org/wp-content/uploads/2020/12/PNEC-2030-Plano-Nacional-Energia-e-Clima.pdf>

²¹ <https://www.statista.com/statistics/792421/cumulative-plant-biomass-capacity-united-kingdom/>

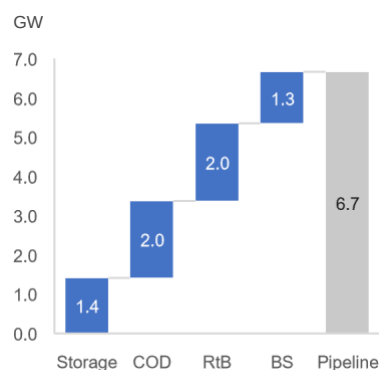


Figure 17: Pipeline breakdown by 2026e.
 Source: Analysts' estimates

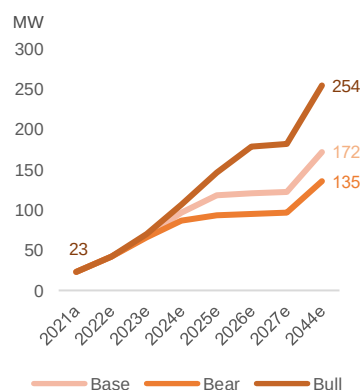


Figure 18: GreenVolt's installed capacity of distributed generation – actual and forecasted. Source: Analysts' estimates

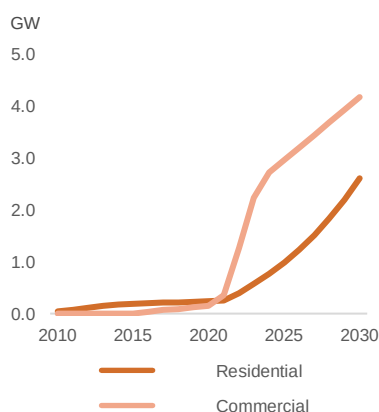


Figure 19: Cumulative PV Solar Installed Capacity in Portugal 2010-2030. Source: Bloomberg

further acquisitions to current installed capacity, we will only forecast the current **pipeline's trajectory into installed capacity per se** (i.e., fully operational).

By **2026e**, we forecast all current pipeline will become **RtB or COD** (within the asset rotation strategy) or **operational** (within the balance sheet strategy), based on the current Company's data on the split between technology, country, and pipeline stage.²² From c.6.7GW of existing pipeline, we forecast 75% will be under the asset rotation strategy, while the remainder will be held on the balance sheet.

Regarding the asset rotation scheme, by **2026e**, we estimate **c.5.4GW** will be either at RtB or COD stages, including the **c.1.4GW** from storage. As for the assets to be held on the balance sheet, we forecast the Company will keep assets in countries on which it has more **expertise** (i.e., Portugal, Poland, and Greece), and keep the already operational solar PV plant in Romania. Thus, by 2023e, we expect c.60MW to be held on balance sheet to be fully operational (i.e., generating cash inflows), and by **2026e**, we expect this number to reach **c.1.3GW**, thus growing by c.22x (Figure 17).

Since the majority of the existing pipeline is early stage, we are not able to draw meaningful conclusions regarding GreenVolt market share in this segment.

Distributed energy generation

As of September 2022, the Company had 33.5MWp of installed capacity and we expect to reach **c.41.9MWp** in the end of 2022, across all scenarios, which represents a growth of 82% from 2021a to 2022e.

From 2023e-2026e, we assume GreenVolt will be able to sustain a high growth rate (though lower than 82%), since (i.) it is the **ramp-up phase of this segment**, and (ii.) the **Iberian countries are GreenVolt's key targets** (Figure 18).

Regarding the latter driver, in Portugal, the low-voltage distribution grid has seen an **expanding connection of small-scale distributed generation and generation intended for self-consumption (both mainly stemming from solar PV)**. From 2015 to 2021, **total installed capacity of small-scale distributed generation grew** from c.18MW to c.61MW (Figure 19). Moreover, in Spain, total installed capacity of both residential and commercial reached c.4.0GW in 2021 (Figure 20). Nonetheless, while sunny Iberian countries seem like it would be innate front-runners in Europe for rooftop solar PV, the region has lingered when compared to other European nations, such as Germany and Italy, which had c.61GW and c.23GW in 2021, respectively.²³ Therefore, **the Iberia presents great**

²² <https://greenvolt.com/wp-content/uploads/2022/11/fsd1239109.pdf>

²³ Bloomberg

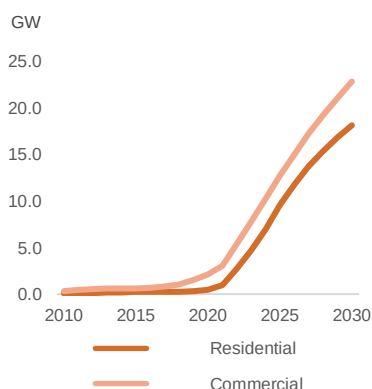


Figure 20: Cumulative PV Solar Installed Capacity in Spain 2010-2030. Source: Bloomberg

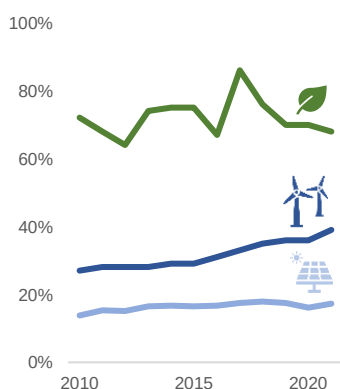


Figure 21: Load Factor by Technology. Source: IRENA

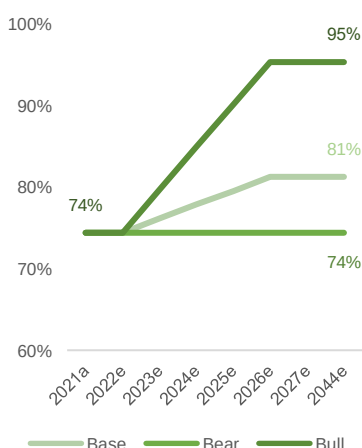


Figure 22: GreenVolt's load factor of biomass plants in Portugal – actual and forecasted. Source: Analysts' estimates

potential, since it has a considerably high horizontal irradiation resource, **yet little current penetration**, which will be favourable for a new-entrant player like GreenVolt, implying a **positive outlook for its market share** in the short term.

After the explicit forecast period, from 2027e onwards, we believe **current government policies and incentives will drive more players to join the self-consumption and energy communities market**. Therefore, the aggregate market is expected to **mature** and follow economic growth, and as such, installed capacity is assumed to grow at the European Union real GDP growth level forecasted for each year.

▪ Load factor

Load factor, also known as capacity factor, offers a glimpse on the reliability of power plants: it measures how often a plant is running at maximum power and it is given by **energy exported over maximum production possible, as per license**.²⁴ This key parameter is crucial to measure plants' functioning, and it **depends mainly on the technology used, the availability of resources, and shutdown periods for maintenance**. According to IRENA, comparing the three technologies, **biomass has the highest load factor**, followed by onshore wind and solar PV. Notwithstanding, these last two followed an upward trend due to technology advancements, while biomass followed a trendless period peaking in 2017 with a capacity factor of 87% (Figure 21).²⁵

Biomass

Load factors for biomass plants are usually very high but come at the expense of regular maintenance work (as boilers often face technical problems), and are highly reliant on continuous access to biomass supply. GreenVolt's power plants in Portugal are powered completely by forestry residues through a long-term agreement by Altri Madeira, and in the UK by urban waste wood through a long-term contract with Esken Renewables. **A clear advantage that contributed to an overall average load factor improvement over the past years.**

In **Portugal**, in the **base** scenario, we assume that **load factor increases by 7pp until 2026e, to c.81%**, since this amount would have been the **average Portuguese load factor as of 2021, if we excluded the Ródão outage**, whose plant load factor would be 71% instead of 40%. In a **bull** scenario, we assume **load factor increases by 22pp until 2026e, to c.96%**, since we consider GreenVolt will be able to apply its **Figueira da Foz II know-how** (i.e., use of

²⁴ <https://www.energy.gov/ne/articles/what-generation-capacity#:~:text=The%20Capacity%20Factor&text=It%20basically%20measures%20how%20often,of%20the%20time%20in%202021.>

²⁵ IRENA - Renewable Power Generation – Costs in 2021

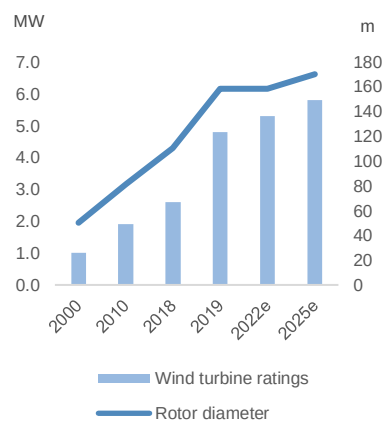


Figure 23: Global average hub height and respective rotor diameter per year. Key characteristics to improve energy yield. Source: IRENA

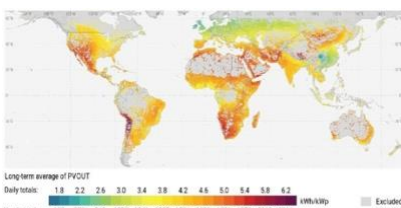


Figure 24: Solar Photovoltaic Power Potential by Country. Source: World Bank

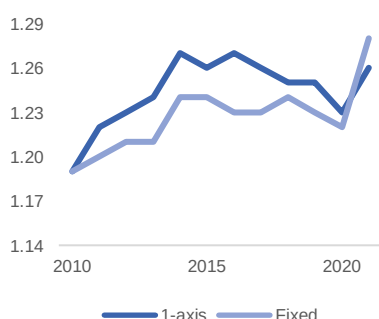


Figure 25: Global average inverter load ratio trend, 2010-2021. Source: IRENA

maximum quality materials, boiler adjustments, among others) in the remaining Portuguese biomass plants, and therefore we expect all load factors to converge to Figueira da Foz II current load factor. In these two scenarios, from 2027e-2044e, load factor remains stable (i.e., growth of 0%). In the **bear** scenario, the no-growth assumption (i.e., no improvement) begins at 2022e (Figure 22).

As for the **UK** load factor (i.e., Tilbury plant), it is already **above the national average load factor** (89% vs. 68%, respectively), therefore **we do not expect improvement (nor deterioration) under in any scenario.**²⁶

In any case, **current load factors (Portugal and UK) are well above the overall worldwide average in 2021 (68%)**, which suggests a favourable outlook for GreenVolt's ability to maintain its competitive advantage going forward and, as a result, its market share.

Solar PV and wind / distributed energy generation

Globally, **onshore wind load factor increased 44% from 2010 to 2021**. It depends essentially on two factors: quality and speed of the air, and turbine technology used. Locations that provide incoming wind consistently, such as mountainous areas, larger and more reliable turbines, along with higher hub-heights and larger rotor diameters, increase load factors (Figure 23).²⁵

The global weighted average **solar PV capacity factor rose from 14% in 2010 to 17% in 2021**. Solar load factor depends essentially on solar resource conditions and technology used. The upward load factor trend is related to (i.) technology advancements, such as the adoption of bifacial technology – i.e., power is produced from both sides of the panel when illuminated, which increases electricity generation – and the use of trackers – which allow the panel to follow the sun's movements and consequently generates more solar power; (ii.) construction of solar PV in sunnier locations, quantified by the global horizontal irradiation (GHI), which changes significantly per country (Figure 24); and (iii.) inverter load ratio (ILR), also known as the DC-to-AC ratio, inverters running close to full capacity lead to a higher load factor (Figure 25).

Overall, we believe **technology will continue to improve**, consequently rising load factors in these two renewable technologies. Although we lack information on current GreenVolt's load factor of these two technologies, we believe it is well positioned to also **surf the technological improvements wave**, thus maintaining competitive advantage going forward (and market share, as a consequence).

²⁶ <https://www.statista.com/statistics/555734/plant-biomass-electricity-load-factor-uk/>

Competitive landscape



Figure 26: Peer Group 1 - companies active in the biomass generation - Albioma, Drax and Ence

In light of the Company's **business mix**, **geographic exposure**, and, more importantly, the **stage of pipeline**, we believe it is **difficult to find a true competitor** within the **market of energy generation from RES**. Nonetheless, GreenVolt may potentially face competition from **several fronts** which can compromise its ability to keep or increase market share.

On one hand, we have **traditional oil & gas players** (e.g., Repsol, Galp, BP) and **integrated utilities** (e.g., Iberdrola, Endesa, EDP, Engie), and on the other, we have **renewable players focused on solar PV and wind** (e.g., EDP Renováveis, Acciona, Orsted) and **traditional biomass players** (e.g., Albioma and Drax). Since GreenVolt positioning lies in between traditional biomass players and pure renewable players, it emerges with **unique** characteristics, such as, **expertise on biomass** (market leader in Portugal), **asset rotation** (due to V-Ridium), and on **developing plants** to RtB or COD.



Figure 27: Peer Group 2 - companies mostly focused on solar PV and wind development - Audax, EDP Renováveis, and Voltalia

As a result, we defined two peer groups, **Peer Group 1** (Figure 26) and **Peer Group 2** (Figure 27), to better understand **if competitors have the required portfolio to capture market share**. We considered firms with market cap between €0.8bn-€2bn and present in Europe (an exception to this is EDP Renováveis, with market cap above €20bn, because we deemed interesting to include a Portuguese peer). Notwithstanding, we note key differences. For example, regarding Albioma, half of its biomass installed capacity is overseas, and Ence and Drax are not pure players (the former has a pulp business and the latter has other generation sources) (Table 1). As for solar PV and wind players, we highlight the discrepancy of installed capacity between GreenVolt and the other peers: GreenVolt has 45MW (solar PV), whereas Audax, the peer with lowest market cap, has 180MW (solar PV and wind) (Table 2).

Biomass

In Portugal, forestry biomass is mainly linked to paper industry. Like GreenVolt, The Navigator Company, the biggest paper firm in Portugal, also takes advantage of pulp facilities for feedstock. In December 2020, it **inaugurated a biomass power station in Figueira da Foz next to its pulp mill**. Nevertheless, given the low growth forecasts of installed capacity for biomass in Portugal, the high barriers to entry associated with high capital costs and the fact that the Navigator Company is not showing signs of future market growth in the biomass industry, we do not believe other players will be able to displace GreenVolt's market share in Portugal.

	Albioma	Drax	Ence
Country			
Market Cap (€m)	1,526	2,165	855
Total IC (MW)	1,029	3,200	266
Share IC Europe	58%	100%	100%

Table 1: Key figures - Peer Group 1. Source: Companies' websites

	Audax	EDPR	Voltaia
Country			
Market Cap (€m)	335	20,479	1,979
Total IC (MW)	180	13,590	1,129
Share IC Europe	89%	42%	23%

Table 2: Key figures - Peer Group 2. Source: Companies' websites

On the other hand, given the **recent Albioma acquisition by KKR**, which may hint further investments in the company, and **Drax's £3bn investment plan** in biomass endeavours, we believe further expansions in Europe would be needed to outdo competition (peers' average installed capacity amounts to c.1.3GW, whereas GreenVolt's amounts to 143MW).^{27,28} Though, in light of the current strategy plan, GreenVolt does not foresee any major expansions, which may help these competitors to capture market share. This, in turn, will be reflected in the overall growth rate for the segment (or lack thereof), which we will further discuss in the RONIC section.

Solar PV and wind / distributed energy generation

There have been several technology improvements in the solar industry which created new products, such as **solar shingles**, **solar windows** and **floating solar panels**. In terms of **innovation** and **use of break-through technologies**, EDP and EDP Renováveis have been front-runners. Regarding the latter, on earlier this year, EDP launched a **floating solar PV power plant** in Alqueva with c.12k photovoltaic panels and an annual production capacity of **c.7.5GWh**.²⁹ Moreover, EDP has a venture capital firm, **EDP Ventures**, specialized in investing in early-stage tech start-ups working in energy, being one of its investment priorities the distributed energy resources. **Offshore wind** has been the latest trend in wind technology, although GreenVolt relies only on onshore, currently. EDP Renováveis through **Ocean Winds**, a joint venture with Engie, is focused on offshore wind farms, with 1.5GW in operation and 16.6GW of total gross capacity.³⁰ On the other hand, for example, **Voltaia** presents an investment plan wherein it allocates c.85% of c.€2.5bn-€3.0bn (which pales in comparison when compared to GreenVolt's CapEx plan until 2026e, as mentioned previously).³¹ Thus, not all competitors will be able to capture market share, but there are major players, as EDP Renováveis, that could pose a serious threat.

- Average selling price

Biomass

GreenVolt's main revenue driver stems from the sale of electricity produced at the plants to the grid, through **PPAs** with guaranteed remuneration: the **FiT** regime in Portugal and the **Renewable Obligation Certificates (ROCs)** in the UK.³²

Power Plant	Price (€/MWh)
Constância	118.3
Figueira da Foz I	120.3
Figueira da Foz II	116.3
Mortágua	132.0
Ródão	122.5

Table 3: GreenVolt FiT Portugal 2021

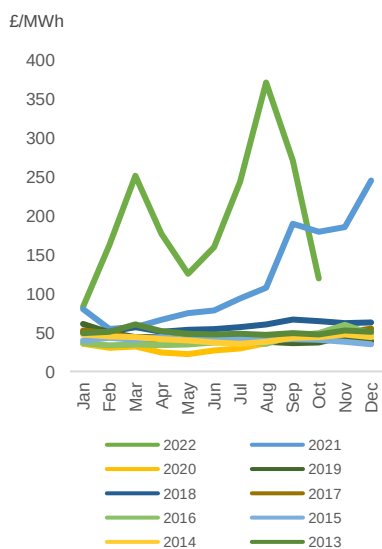


Figure 28: N2ex Day Ahead Auction Prices in the UK 2013-2022. Source: Nordpool

²⁷ https://media.kkr.com/rss-feed/news-release/?news_id=257162f3-2f33-4968-acae-5680c956238a&type=1

²⁸ <https://www.drax.com/wp-content/uploads/2022/03/Drax-2021-FYR-Analyst-Presentation-website.pdf>

²⁹ <https://www.edp.com/en/innovation/floating-solar-pannels>

³⁰ <https://www.oceanwinds.com/ocean-winds-in-numbers/>

³¹ <https://www.voltaia.com/static-files/c3f5feed-9ce7-4f6c-8d9b-e6b5a32e0f34>

³² Additionally, in the UK, there is also a variable component that changes according to the evolution of the electricity price (Brown Power)

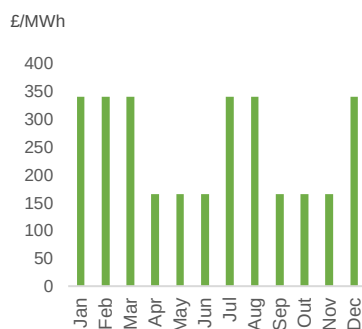


Figure 29: Monthly selling prices for 2023 and 2024 for the UK. Source: Analysts' estimates

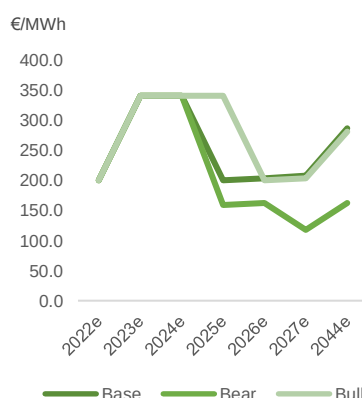


Figure 30: GreenVolt's average prices of biomass in the UK – forecasted. Source: Analysts' estimates

In Portugal, in the **base** and **bull** scenarios, **average selling price (€/MWh) from 2022e-2026e corresponds to the average FiT during 2021 (€121.9/MWh)** (Table 3); from **2027e-2044e, we expect the average selling price to converge to the national average indicative tariff (€119.0/MWh)**.^{33,34}

In both these scenarios, we assume FiT regimes applicable to each plant are extended, in light of the **recent 17-year extension for the Mortágua plant**. For the **bear** scenario, we **do not expect FiT extensions** and, as such, GreenVolt will only be entitled to the fixed remuneration until FiT expiry; **upon termination, the Company becomes subject to market prices**. The first FiT expiry occurs in 2031e, so in **2032e we considered an average selling price of €60.5/MWh**, which corresponds to the EEX Spanish Power Futures, provided by the European Energy Exchange, with a predetermined future date of 2032.³⁵ We assumed this futures contract would be a good proxy for Portugal, since the Spanish market is close to the Portuguese one (MIBEL). Notwithstanding, we understand this is a non-arbitrage price and that futures contracts are mainly used to hedge risk but, as the Federal Reserve Bank of St. Louis pointed out, they may have a side advantage of often suggesting good price forecasts.³⁶ From that year onwards, the average selling price will grow at the forecasted inflation level.

In the UK, according to the Office of Gas and Electricity Markets (Ofgem), the Renewables Obligation (RO) is a support mechanism for large-scale renewable electricity projects, and attributes an obligation on UK electricity suppliers to provide an increasing proportion of electricity they supply from renewable sources; failure to meet this obligation implies a buy-out payment.³⁷ Since there is no historical data on GreenVolt's ability (or lack thereof) to meet its obligations, we assume a **no-failure case**. Thus, the market price is the only relevant driver for the average selling price (Figure 28). For **2022e, we assume the average N2ex Day Ahead Auction Prices in the UK, from January until November 2022 (£199.7/MWh) under all scenarios**.³⁸

Given the current unprecedented and volatile times, the UK government set an Energy Price Guarantee (effective as of October 2022) with a **cap of £340/MWh until April 2024e**.³⁹ It is challenging to forecast energy prices during normal times, but it is even more so during an energy crisis. Chris O'Shea, from Centrica, owner

³³ <https://www.dgeg.gov.pt/pt/areas-setoriais/energia/energia-eletrica/producao-de-energia-eletrica/producao-potencia-instalada-superior-a-1-mva/remuneracao-das-energias-renovaveis/>

³⁴ For a coefficient Z=9.6

³⁵ <https://www.eex.com/en/market-data/power/futures/#%7B%22snippetpicker%22%3A%2234%22%7D> – as of November 4, 2022

³⁶ <https://www.stlouisfed.org/publications/regional-economist/january-2002/the-futures-market-as-forecasting-tool-an-imperfect-crystal-ball#:~:text=Although%20futures%20contracts%20primarily%20exist,also%20produce%20good%20price%20forecasts.>

³⁷ <https://www.ofgem.gov.uk/environmental-and-social-schemes/renewables-obligation-ro>

³⁸ <https://www.nordpoolgroup.com/en/Market-data/1/GB/Auction-prices/UK/monthly/?view=table>

³⁹ <https://www.ofgem.gov.uk/information-consumers/energy-advice-households/check-if-energy-price-cap-affects-you>

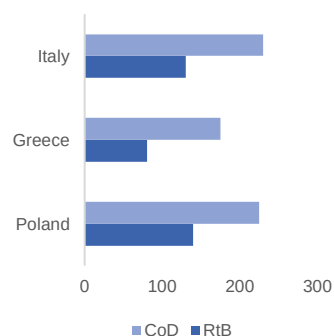


Figure 31: Solar - Project Exit Value per MW per country. Source: GreenVolt

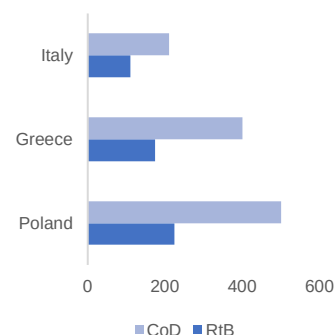


Figure 32: Wind - Project Exit Value per MW per country. Source: GreenVolt

Country	Solar	Wind
Portugal ¹	38.0	n.a.
Poland ²	71.4	48.3
Greece ²	54.0	63.0
Romania ³	37.0	n.a.

Table 4: Remunerations per country (€/MWh). Note: ¹ – PPA; ² – CfD; ³ – Market Price. Source: Roedl, Balkan Green Energies, GreenVolt Capital Increase Prospectus

of British Gas, said that it is expected that for the next two years, prices will continue at an all-time high.⁴⁰ As a result, for all scenarios, in 2023e and 2024e, we considered that the cap will be met on the summer and winter months and, for the remaining, we used the average spring and autumn months of 2022a (Figure 29). Only in the bull scenario, we considered that the cap will be extended until 2025e.

From 2027e onwards, we considered a **return to pre-Ukraine war electricity price levels**, since current forecasts and research do not suggest otherwise. For the base and bull scenarios, we assumed the same selling price as for 2022e, which will be indexed to the forecasted inflation for each year going forward. As for the more severe scenario, we believe selling prices will return to **N2ex Day Ahead Auction Prices in the UK for 2021 (£117.4/MWh)** (Figure 30).

Solar PV and wind

Revenue streams from this business segment are threefold: (i.) provision of accounting, administrative and asset management services, (ii.) **sale of solar and wind energy projects, in a RtB or COD phase**, and (iii.) **fixed remuneration schemes from the plants developed to own**. We consider the first stream to be non-meaningful, when compared to the other two.

Regarding the asset rotation strategy, GreenVolt recently executed its **first sale solar PV and wind agreement** with Iberdrola, for **€155m**, of a portfolio of assets (2 wind farms and 6 solar PV farms) under construction in Poland, with an installed capacity of 98MW.^{41,42} Since this is the first sale, we do not have much visibility to extrapolate forecasts. However, the Company shared **value creation schemes** (detailing development costs and exit values), for **both stages** (RtB and COD) and for **each technology** (solar PV and wind), for Poland, Greece, and Italy.⁴³ Hence, we extrapolated from this available information **averages of selling prices (i.e., the exit values)** for each technology and each stage, as seen in Figure 31 and Figure 32.

On the other hand, given its deep knowledge and industry expertise, GreenVolt wants to maintain a portfolio of contracted assets in its balance sheet. These projects have remunerations based on **PPA or Contract-for-Difference (CfD)**, which mitigates the risk and hedge against energy market price fluctuations.⁴⁴ To

⁴⁰ <https://bionic.co.uk/blog/noticed-energy-prices-have-fallen-over-past-few-months/>

⁴¹ All the assets in the portfolio are covered by the PPA signed with T-Mobile Polska

⁴² <https://econews.pt/2022/08/01/greenvolt-offloads-solar-wind-farms-in-poland-to-iberdrola-for-e155m/>

⁴³ https://greenvolt.com/wp-content/uploads/2022/06/pdf_en_20.pdf

⁴⁴ CfD is based on a difference between the market price and an agreed strike price. If the latter is higher than the former, the CfD counterparty must pay the renewable producer the difference (IEA, 2019)

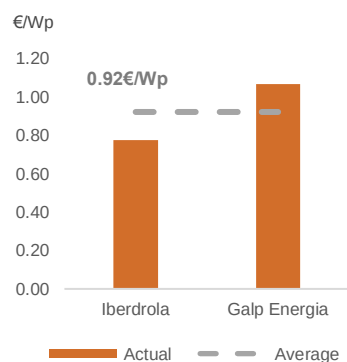


Figure 33: Distributed Generation - B2B - Selling Price (Peer analysis). Source: Companies' website

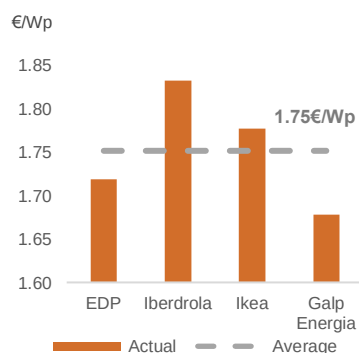


Figure 34: Distributed Generation - B2C - Selling Price (Peer analysis). Source: Companies' website

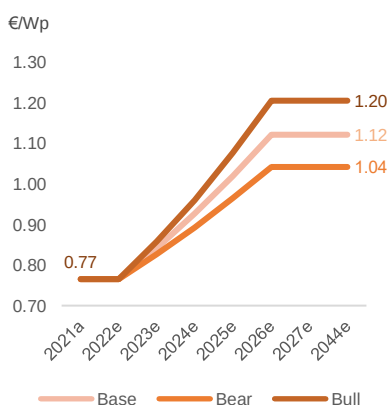


Figure 35: GreenVolt's average prices of distributed generation – actual and forecasted. Source: Analysts' estimates

estimate the average selling price, we checked in-place remuneration schemes for each country and for each technology, which are summarized in Table 4.

Distributed energy generation

As for distributed generation, GreenVolt's revenues mainly come from **the installation and maintenance of distributed solar energy production units (B2B and B2C)**, and also from the development and financing of projects to improve energy efficiency through solar energy.

Regarding the former, due to the lack of information concerning prices set by the Company, **we extrapolated the average selling price from GreenVolt's revenues and installed capacity as of September 2022: €0.77/Wp or c.€765k/MWp.**⁴⁵

To assess the reasonability of such extrapolation, we conducted a **peer analysis in the Portuguese market, in both market segments**. For the **B2B segment**, we considered prices from Iberdrola and Galp Energia, resulting in an average of **€0.92/Wp** (Figure 33). For the **B2C segment**, we considered comparable offerings from EDP, Iberdrola, Ikea, and Galp Energia, resulting in an average of **€1.75/Wp** (Figure 34). Since we lack visibility regarding the weight of each segment in GreenVolt's offerings, we applied an equal weighted average, resulting in **€1.34/Wp or €1.340k/MWp**. We concluded the extrapolation is fair, given GreenVolt is a new entrant in the installation of solar panels market (i.e., setting low prices to penetrate the market), and **we set the average selling price for 2022e as c.€765k/MWp**.

In all cases, we assume GreenVolt will be able to set higher prices in the short-term (2023e-2026e), as it gains traction, **approaching the overall market**, at different growth rates, but never surpassing it. Selling prices will grow by 10pp, 8pp, and 12pp each year in the short-term, in the base, bear and bull scenarios, respectively. Thus, average selling prices by 2026e are expected to be **€1.120k/MWp, €1.040k/MWp, and €1.204k/MWp**, in the base, bear, and bull scenarios, respectively. From 2027e onwards, prices will sustain in all scenarios (Figure 35).

Regulatory framework

The **Paris Climate Agreement**, an international treaty on climate change, whose chief goal is to limit global warming, for which the transition towards renewables such as solar PV and wind play a critical role, has already yielded positive outcomes. After more than 5 years after the Agreement, **c.70% of the countries**

⁴⁵ <https://greenvolt.pt/wp-content/uploads/2022/09/Greenvolt2Q22PT.pdf>

**Main regulatory framework
impacting GreenVolt**

Paris Climate Agreement
European Green Deal
Clean Energy Package
"Plano Nacional Integrado Energia Clima 2030"
Spanish National Integrated Energy and Climate Plan 2021-2030
European Law on Climate
FiT for 55
Climate Framework Law
REPowerEU Plan

Table 5: Energy regulations impacting GreenVolt

involved disclosed quantified renewable energy targets, which can be read as a driver to foster energy demand for these technologies.^{46,47} Additionally, since the introduction of the **EU Renewable Energy Directive** (2009/28/EC) in 2009, **the use of renewables has been increasing yearly**, reaching more than 22% in 2020, which puts the EU in a leading position for renewables technology improvement and deployment.⁴⁸ The European Commission has recently presented the **REPowerEU Plan**, its reaction to the adversities and global energy market disruption caused by Russia's war on Ukraine. Measures foreseen in the plan include **the broadening of energy supplies, and quicker roll-out of renewable energy to replace fossil fuels** in homes, industry, and power generation.⁴⁹ Many other agreements and laws have been passed and enacted towards a **future where RES play a far greater role than today** (Table 5).

Regarding **incentive schemes**, we acknowledge that the transition towards climate neutrality requires a wider policy context that integrates a broad range of instruments, such as **remuneration mechanisms** and **subsidies**. Concerning the latter, energy subsidies in the EU reached an amount to c.€180bn in 2019. Despite EU's pledges to phase out fossil fuel subsidies, research shows they are not decreasing adequately. Nonetheless, **renewable energy subsidies amounted to c.€80bn, an improvement of 8% since 2015**.⁵⁰ Regarding the former, mechanisms as FiT and ROC play a key role in incentivizing energy generation through RES, since energy markets alone cannot bring the targeted level of renewables in the EU, requiring **national governments to enact support schemes to foster increased investment in renewable energy**.⁵¹

Overall, between agreements, laws and incentive schemes, we believe the current state of affairs to be very bright in what RES are concerned, which impacts GreenVolt positively.

- Other macroeconomic factors: exchange rates, inflation and real GDP growth

GreenVolt has **pan-European operations** and is consequently subject to **exchange rate fluctuations**, so **foreign-currency risk** must be incorporated in the valuation.⁵² In fact, in the event of a currency exchange rate adversely changing, it may potentially lead to **(economic) value destruction**. In particular,

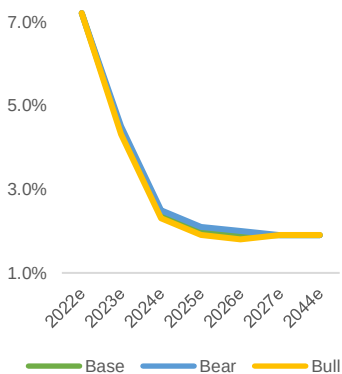


Figure 36: Inflation forecast. Source: IMF

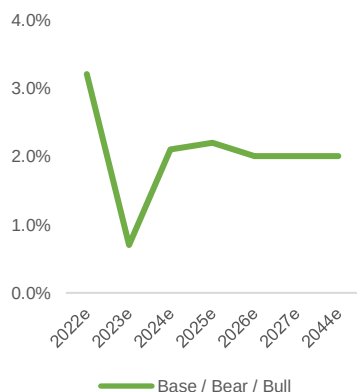


Figure 37: EU real GDP growth forecast. Source: IMF

⁴⁶ <https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement>

⁴⁷ <https://www.irena.org/publications/2020/Dec/Renewable-energy-and-climate-pledges>

⁴⁸ https://energy.ec.europa.eu/topics/renewable-energy/renewable-energy-directive-targets-and-rules/renewable-energy-directive_en#:~:text=The%20directive%20establishes%20a%20new,the%2020%25%20target%20for%202020.

⁴⁹ https://ec.europa.eu/commission/presscorner/detail/en/IP_22_3131

⁵⁰ https://www.ffms.pt/sites/default/files/2022-07/state_of_the_energy_union_report_2021.pdf

⁵¹ https://energy.ec.europa.eu/topics/renewable-energy/financing/support-schemes_en

⁵² Valuation – measuring and managing the value of companies, McKinsey, 6th edition

we highlight the operations in the **UK**, through the **Tilbury plant**, and in **Poland**, through **V-Ridium**, which represent a high percentage of EBITDA and pipeline, respectively. Therefore, to convert cashflow projections we will use the **forward-rate method**, as suggested by McKinsey: we will forecast foreign cash flows in the foreign currency, and convert these into euros using the relevant forward exchange rates. We expect the **Pound Sterling (GBP) forward exchange rate to EUR** and **Polish Zloty (PLN) forward exchange rate to EUR**, for **2022e**, to be **1.14** and **0.21**, respectively, based on current forwards, as per data from the Financial Times.^{53,54} Since we lack visibility for the next years, 2022e will be the best forecasted year, and from 2023e-2024e, we assume the same value.

Based on IMF predictions for advanced economies, **we forecast inflation to be at 7.2% in 2022e in all scenarios**.⁵⁵ For **2023e-2026e**, in the **base** case we continued to assume IMF predictions; our forecasts are **1pp above and 1pp below IMF's** in the **bear** and **bull** scenarios, respectively. **From 2027e onwards, we assume 1.9% for each forecasted year, in all scenarios**. In this later stage (i.e., 2027e-2044e), we believe the Company will be **more mature** and thus **able to increase selling prices at the inflation level**, enough to overcome higher capital costs as well as operating costs (Figure 36).

For the EU real GDP growth, and since we only consider this factor to be relevant in a more mature stage (i.e., 2027e-2044e), when we believe **GreenVolt will be able to grow quantities** (e.g., DG installed capacity) **at the economy level**, we forecasted this growth to be equal to the one forecasted by IMF for **2026e (2%)** (Figure 37).⁵⁶

Industry cost drivers: analysis and forecasts

▪ Feedstock costs

Biomass feedstocks are materials used to produce energy and it includes plants, wood and waste. The **availability** and **quality of feedstock** is crucial for the daily operations of biomass power plants. The presence of sand and water negatively impacts plants' load factor and increases the probability of equipment failure. Feedstock costs depends on the type of feedstock use, but also on how its harvested, transported and stored. The prices of pellets and woodchips are quoted in Europe by ENDEX and PIX.⁵⁷

⁵³ <https://markets.ft.com/data/currencies/tearsheet/summary?s=GBPEUR>

⁵⁴ <https://markets.ft.com/data/currencies/tearsheet/summary?s=PLNEUR>

⁵⁵ <https://www.imf.org/external/datamapper/PCPIPCH@WEO/ADVEC>

⁵⁶ https://www.imf.org/external/datamapper/NGDP_RPCH@WEO/EU

⁵⁷ IRENA - Biomass for Power Generation - 2012

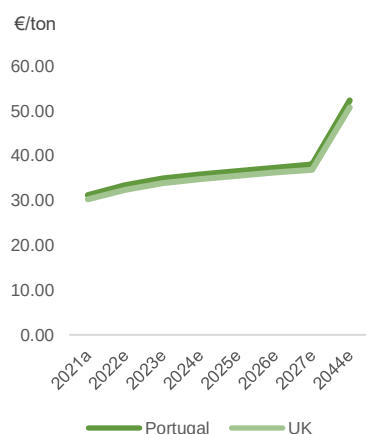


Figure 38: Market Price - Feedstock - actual and forecasted – Base case. Source: Analysts' estimates

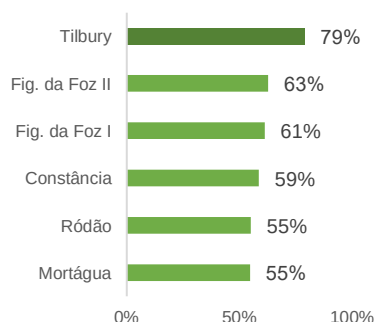


Figure 39: Operational Margins per biomass plant 2021. Analysts' estimates

GreenVolt has a **long-term supply agreement with Altri**, under which the latter agreed to deliver a specific quantity and quality of feedstock on an agreed-upon delivery date. Altri directly supplies from its own paper pulp facilities production process between 30% to 40% of GreenVolt's biomass needs. **The price is fixed at €27.6/ton for the duration of the contract which is identical to the FiT of each plant.** The remaining is also managed by Altri, either from Altri's own entities or third parties. In this case, the price is subject to **merchant prices**. Using past data, we estimated that the price paid for feedstock is **€31.2/ton**, which results in an overall price for Portugal of **€29.9/ton**. On the following years, the price is indexed to inflation. As for the Tilbury plant, it has a **long-term fuel supply agreement with Esken Renewables until 2037**. The price paid is indexed at Retail Price Index ("RPI") and in 2021 it was **€30.2/ton**. As for logistics, all feedstock supplied comes from areas close to the plants, as a result transportation costs, and risk of feedstock deterioration are minimized (Figure 38). As discussed, these contracts provide GreenVolt with a competitive edge which is translated into operating margins that go in line with the industry average (c.63%), according to NYU Stern (Figure 39).⁵⁸

▪ Instalment costs

For **distributed energy generation**, **copper** and **aluminium** are essential for solar cells, and **as the deployment of solar module increases**, these minerals and metals become of chief importance when forecasting installation costs (Table 6). As a result, we added them in our valuation using future agreements. In the short-term, we expect the costs of distributed generation to **increase slightly due to rising commodities prices, mainly aluminium, and supply chain disruptions**.

▪ Operational Expenditure (OpEx)

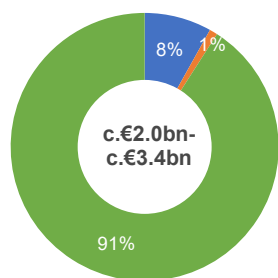
Regarding **biomass**, we highlight OpEx costs such as operation and maintenance ("O&M"), suppliers, and insurance – forecasted as percentage of sales. For **DG**, innovation and technological advancements result in solar modules that are **optimised to reduce O&M costs**. Schedule inspections are necessary to ensure cleaning and repairs. GreenVolt considers the **maintenance of solar PV panels to be minimal and it is already incorporated in their selling price**. The other costs are forecasted as a percentage of sales, as well.

▪ Capital Expenditure (CapEx)

	Copper	Nickel	REEs	Chromium	Zinc	Aluminium
Solar PV	●	○	○	○	○	●
Wind	●	○	○	○	○	○
Bioenergy	○	○	○	○	○	○

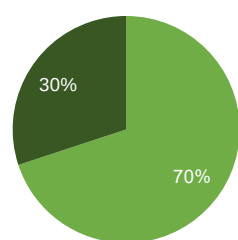
Table 6: Critical mineral needs for clean energy technologies. Note: Shading indicates the relative importance of minerals for a particular clean energy technology (Dark dot = high; grey dot = moderate; white dot = low). Source: IEA

⁵⁸ https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/margin.html



■ Biomass ■ DG ■ Solar PV and wind

Figure 40: CapEx forecasted investment plan 2022e-2026e, per segment. Source: Analysts' estimates.



■ Maintenance ■ Expansion

Figure 41: CapEx forecasted breakdown 2022e-2026e, for the biomass segment. Source: Analysts' estimates.

Biomass

The major components of CapEx in biomass plants are **planning, engineering, building expenses, fuel handling and preparation machinery, and other equipment**.²⁵ Biomass power plants and its equipment requires **ongoing investments** to be kept in proper shape. In fact, maintenance CapEx makes up most of the total CapEx (70%) between 2022e-2026e, and as result, expansion CapEx contributes to only c.30%, which is in line with our assumption of no major expansions forecasted (Figure 41).

To forecast total property plant and equipment ("PPE"), intangible assets ("IA"), and goodwill, we projected **PPE and IA based on MW**, following the same assumptions used in the revenue forecast about the energy crises. During those years, we expect PPE/MW and IA/MW to **rise due to high commodities prices and supply chain risks**. Afterwards, the value maintains constant. For **goodwill**, we forecasted 2022e based on HY 2022 reported values (to account for acquisitions during the period), and from 2023 onwards, we kept the value constant, since we are not factoring in further acquisitions. We set **maintenance CapEx** to equal **amortization and depreciation of each year**, and then we set **total CapEx as the difference of total PPE, IA, and goodwill from one year to the previous, plus maintenance CapEx**. The difference between total CapEx and maintenance CapEx yields **expansion CapEx**, needed to account for possible greenfield investments.

Solar PV and wind

As we will delve in the EV/MW model section ahead, for the asset rotation pipeline, we considered the **value creation schemes** mentioned before for the cashflows projection. For the develop to own pipeline, we factor in both **installation CapEx and maintenance CapEx** into the cash outflows projection. For the **installation costs**, we considered c.€754k/MWh and c.€1,166k/MWh, for solar PV and wind farms, respectively, based on IRENA's dataset for 2021. These costs will occur from 2023e-2026e (after which we do not factor in further expansions, and we assume all pipeline will become operational). Moreover, we assumed also **maintenance CapEx** from 2024e-2026e, as a **percentage of installation costs** (3%). From 2027e onwards, we factor in solely maintenance CapEx, as we deem maintenance of utmost relevance for the plants to generate cash inflows throughout time (Figure 42). For the Romanian solar park in operation, we considered only maintenance CapEx for all forecasted periods.

Distributed generation

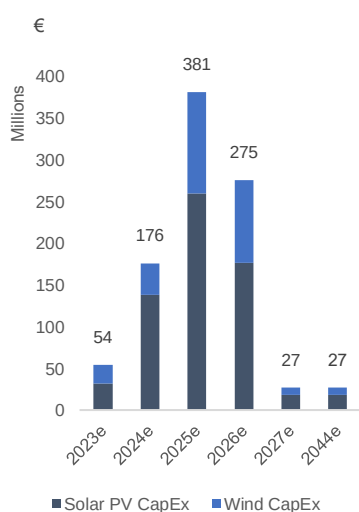


Figure 42: CapEx forecasted breakdown per technology (develop to own strategy). Source: Analysts' estimates.

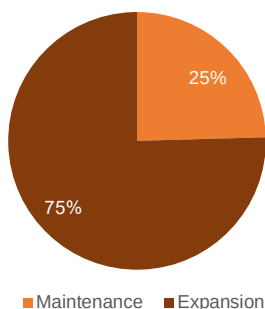


Figure 43: CapEx forecasted breakdown 2022e-2026, for the distributed energy generation segment. Source: Analysts' estimates

PPE/MW and IA/MW are forecasted to growth by 15% each year between 2022e-2026e to factor in the **ramp-up phase** of DG. As expected, expansion CapEx takes up to c.75% of total expenditures (Figure 43).

Risks

Biomass supply chain risk

Overall, the cost of biomass represented 42% of the electricity sales in 2020 and 32% of the electricity sales in 2021. The **long-term contract** with Altri Madeira in Portugal and Esken Renewables in UK **minimizes the risk of supply procurement shortage**. On one hand, **part of the contracts is subject to market prices fluctuations**, on the other, the biomass cost is calculated based on achieved generation output and FiT has a CPI passthrough, which provides **GreenVolt a hedge regarding fluctuations**. Notwithstanding, GreenVolt cannot set aside the impact of biomass supply shortages and market prices volatility.

Pipeline mortality rate risk

The majority of the projects are at an advance and early stage (before licencing): c.96% of 6.7GW (total pipeline). **Project delays** and an **increase in mortality rate** might **decrease market sentiment** and consequently impact negatively the firm's valuation. Nonetheless, given V-Ridium **experienced team and brand recognized** we believe the risk is minimized.

Windfall tax risk

GreenVolt's business is also affected by other taxes and levies. Recently, measures related to a new temporary windfall tax (levied on unexpected large profits) have been announced throughout Europe. In the UK, where GreenVolt is currently subject to **high wholesale prices**, while the measure is still under discussion, we cannot anticipate unfavourable impacts from this tax, due to lack of government guidance.⁵⁹

Valuation

SOTP model

⁵⁹ The Romanian windfall tax will be in force until August 2023 which we consider to yield an immaterial impact in GreenVolt

Bearing in mind GreenVolt's updated strategy plan, it is clear how different the underlying dynamics for each segment are. Thus, we believe a **Sum-of-the-Parts ("SOTP")** approach is better, especially in what concerns **cashflow stability**.

The **biomass segment is expected to maintain stable cashflows with long visibility**, due to in-place remuneration schemes. Therefore, we assume a discounted cash flow (**DCF**) model is the most suitable. Although the **DG segment is in its ramp-up phase** (thus we lack visibility), we believe **DCF** is also the best valuation method. For these two segments, we considered an **explicit forecast between 2022e-2026e**, to match the current company's strategy plan, and the **implicit forecast period will last from 2027e until 2044e, which coincides with the termination of Figueira da Foz FiT regime**. As explained, given the recent Mortágua plant FiT 17 years extension (i.e., FiT expiry in 2041 instead of 2024), and the special characteristics of the biomass business in Portugal, we believe **further extensions are likely to follow in the remaining Portuguese plants**.

Regarding the **solar PV and wind segment**, since current pipeline is mostly in an early stage (mortality rate is an issue), we created an **EV/MW model** from two different standpoints: method 1, where we **detailed the asset rotation/develop-to-own strategy and respective expected cashflows, deriving value through NPV**, and method 2, where we cross-checked value by **considering both the different stages of pipeline and precedent transactions**.

Net Working Capital and Invested Capital analysis

For **Net Working Capital (NWC) requirements**, most forecasts are based on **revenues** (operating cash, assets associated with contracts, and other payables). For inventories, trade receivables, and trade payables, we based our expectations on **activity ratios' analysis** (holding period, collection period, and payable period, respectively). Indeed, comparing to Peer Group 1, GreenVolt is close to the average in each ratio, so we forecasted on an average historical basis, for the biomass segment. As for DG, due to lack of historical information, and since Peer Group 2 leans more towards the solar PV and wind reality than distributed generation itself, we forecasted these ratios based on 2021a.

For Invested Capital (IC), the most relevant captions (PPE and IA) were already discussed in the CapEx chapter.

Biomass

In the base scenario, NWC will experience a **high-growth period between 2021a and 2022e (59%)** mostly due to the increase in revenues, in light of the **sky-rocketing electricity prices in the UK**, as mentioned earlier. In 2025e, since we

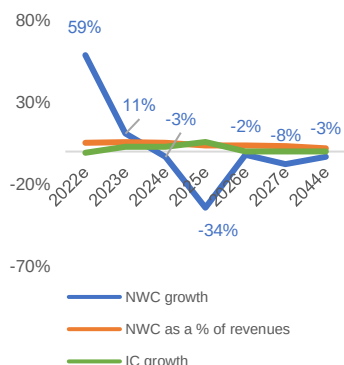


Figure 44: NWC and IC analysis for the distributed energy generation segment under the base scenario. Source: Analysts' estimates

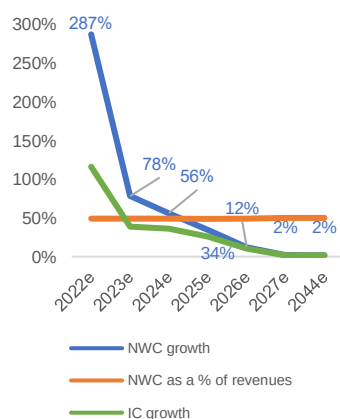


Figure 45: NWC and IC analysis for the biomass segment under the base scenario. Source: Analysts' estimates

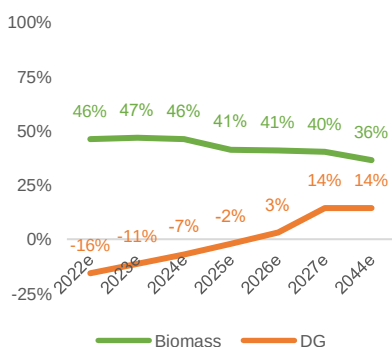


Figure 46: EBITDA margin for the biomass and distributed energy generation segments under the base scenario. Source: Analysts' estimates

assume electricity prices will return to 2022e levels, NWC will decrease substantially (by c.34%). After that, it will slowly decrease, as electricity prices return to normalcy (Figure 44). Overall, forecasted NWC requirements as a percentage of revenues are expected to fall in line with J.P. Morgan's Working Capital Index (c.5%) and Albioma's as of 2021 (c.8%), which translates GreenVolt's ability to meet short-term obligations with short-term in line with the industry.⁶⁰

Regarding Invested Capital (IC), we highlight the **6% increase in 2025e** (base scenario), due to **the new Mortágua plant**, which is bound to **increase PPE by c.7%** in that year. From **2027e** onwards, IC is also expected to **decrease slowly each year** – which translates the **lower investment expected** for this segment going forward (Figure 44). In the bear and bull scenarios the same rationale applies but factoring in the assumptions laid out previously.

Distributed energy generation

In this segment, both NWC and IC present a high growth period during 2022e-2026e (ramp up phase) compared with the implicit forecast period (Figure 45). In particular, we highlight 2022e, when PPE increases by c.2.1x.

Forecasted margins

The biomass segment, which is still the **main cashflow generator within the Company**, is expected to **become less pronounced in terms of EBITDA contribution**, in line with the updated strategy defined. In fact, by **2027e**, which is when DG has its **first positive EBITDA contribution under all scenarios**, this segment will contribute by around 23% to EBITDA, whereas biomass will contribute by 77%.⁶¹

Nonetheless, biomass is expected to present a **higher EBITDA margin** when compared to DG going forward (e.g., in 2027e, c.40% vs. c.14%, respectively, in the base case), since the former benefits from long term fixed remuneration schemes (Figure 46). We note that GreenVolt's biomass EBITDA margin and EBIT margin going forward are **above the peers**, which depicts its deep expertise in the segment.

Discount rate

⁶⁰ <https://www.jpmorgan.com/content/dam/jpm/treasury-services/documents/working-capital-report-2022.pdf>

⁶¹ This analysis excludes the solar PV and wind segment, since we do not have the respective EBITDA breakdown. If we were to include this segment, we would expect an EBITDA split that follows GreenVolt's guidance (15% for distributed energy generation, 30% for biomass, and 55% for solar PV and wind). Having this guidance in mind, if we exclude solar PV and wind, EBITDA contribution would be 33% and 67% for distributed generation and biomass, respectively, which is in line with our 25%-75% split

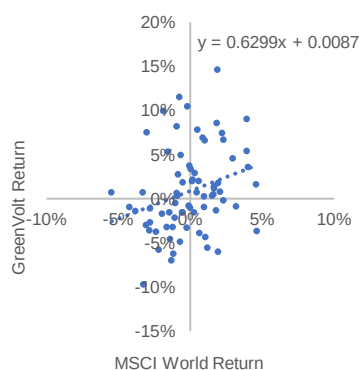


Figure 47: Scatterplot of Weekly Returns for GreenVolt vs the MSCI World Index, 2021-2022. Source: Bloomberg

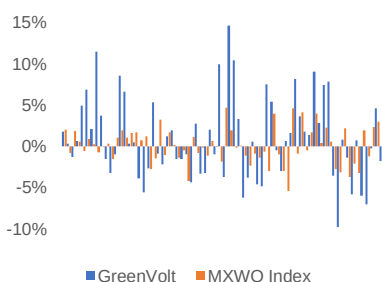


Figure 48: Weekly returns for GreenVolt Stock and for the MSCI World Index, 2021-2022. Source: Bloomberg

We calculated **two** WACCs to track the different characteristics of the biomass segment and solar PV and wind/DG segment. The WACC three primary factors are: **cost of equity**, **cost of debt** and **capital structure**.

Cost of Equity

We used the Capital Asset Pricing Model (“CAPM”) to compute the cost of equity. The components used are **risk free**, **market risk premium (“MRP”)** and **beta**.

The proxy used as **risk-free** was the **German Bund 10Y Yield** as it has a **low default risk** and the **same currency as the GreenVolt’s cash flows**. Even though it is not **the same duration of the cash flows**, this yield is **more tradable than the Bund 30Y Yield**, therefore the **best proxy for the risk-free**. As for **MRP**, after carefully analysing the methodology, we used 6.0%, as suggest by KPMG.⁶²

Despite, the best strategy to avoid systematic biases and noisiness is to use **monthly returns**, GreenVolt’s stock only went public on **15 July of 2021**, therefore **it would not be possible to have sixty data points for raw regressions**, as a result, we opted to choose **weekly returns**. Nevertheless, as GreenVolt stock is traded on a continuous basis, we do not expect the beta to be highly impacted between choosing weekly or monthly data. We graphed the rolling betas to search for patterns in the stock’s risk (Figure 48).

To forecast the equity beta in the future, we used historical returns as we believe they will remain stable over time. **We estimated the raw beta of GreenVolt by regressing its returns against the Morgan Stanley Capital International (“MSCI”) World Index (proxy of market return).**⁶³ It resulted in a beta of 0.63 with a confidence interval of [0.12, 1.14]. Given the broad range, and the fact that individual firm betas can be profoundly influenced by unrepeatable events like financial crises, we believe the **best strategy to improve precision is to create an industry beta**. As a result, we estimated the raw betas of each peer groups following the same technique abovementioned. Afterwards, we applied the **smoothing process** used by Bloomberg, based on the evidence that betas converge to 1 in the long term.

A company’s beta is a function of its operating and financial risk. To compare betas of different firms across an industry, we need to **offset the effect of leverage embedded by the financial risk**. Considering Franco Modigliani and Merton Miller work, we unlevered the raw adjusted betas, taking into account that the default risk in this industry is low, therefore we considered the beta of debt as

⁶² Equity Market Risk Premium – Research Summary – KPMG

⁶³ For the sake simplicity, we used returns instead of excess returns to forecast the betas as there is low meaningful difference (Credit Suisse, 2013)

WACC PEER GROUP 1	
Cost of Equity	6.45%
Cost of Debt	5.11%
Marginal Tax Rate	21.00%
E/EV	69.23%
D/EV	30.77%
WACC	5.71%

Table 7: WACC – Biomass. Source: Analysts' estimates

WACC PEER GROUP 2	
Cost of Equity	5.78%
Cost of Debt	5.11%
Marginal Tax Rate	21.00%
E/EV	69.23%
D/EV	30.77%
WACC	5.24%

Table 8: WACC - Solar, Wind and DG. Source: Analysts' estimates

zero.⁶⁴ We removed outliers such as Albioma in Peer Group 1 which is driven by anomalies we do not believe will persist in time.

Afterwards, we **relevered the beta using GreenVolt's target capital structure and the median unlevered industry beta**, which resulted in a beta of 0.75 for Peer Group 1 and 0.64 for Peer Group 2. Applying the CAPM formula, we ended up with a **cost of equity of 6.45% and 5.78%**.

Cost of Debt

We performed **credit risk analysis** following two approaches: **(i.) a synthetic credit rating based on interest coverage ratio**, and **(ii.) a credit scoring model - Z-score** developed by Altman (2006).

Looking at 2021a values, GreenVolt has an interest coverage ratio of 5.9x. According to NYU Stern, this interest coverage ratio value is linked to a **synthetic rating of A-**. The second method used was developed by **Altman**: a model that matches Z-scores and observed ratings. GreenVolt **Z-score is 5.9**, which is considered a **BBB- rating**. From these two approaches, GreenVolt is considered an **investment grade company**, which goes in line with the first **credit rating attributed to GreenVolt on 2nd September 2022** by Ethifinance.

As a result, the probability of default is immaterial, thereby a good proxy for cost of debt is the **yield to maturity of the Company longest bond, resulting in a cost of debt of 5.11%**. The cost of debt was calculated for the Company as a whole, therefore as we do not have evidence for it to be different between peer group, we will use the same cost of debt for both WACC.

Regarding the **marginal tax rate**, the average of all countries GreenVolt is present is 20%. For the sake of simplicity and given that it is not that different from Portugal's income tax rate, we used 21% as a marginal tax rate for both groups, which is consistent with our valuation model. The **after-tax cost of debt is 4.03%**.

WACC

Based on the **cost of equity** and **after-tax cost of debt** computed above, we estimated an WACC of **5.71%** and **5.24%** for Peer Group 1 (Table 7) and Peer Group 2 (Table 8), respectively, using GreenVolt's **current capital structure** (D/E of 30.7%), since we assume this to be the Company's **target ratio** going forward, as we have no evidence to think otherwise.

⁶⁴ While we acknowledge it is a simplification method (according to McKinsey), we do not believe it will have a meaningful impact to consider a beta debt of zero

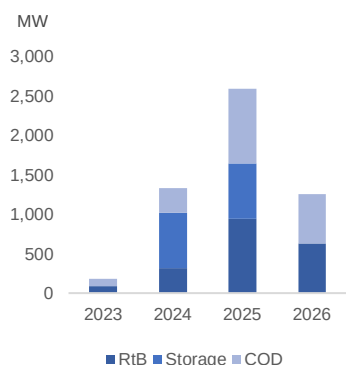


Figure 49: Method 1 - Asset Rotation - Split per technology). Success rate 100%. Source: Analysts' estimates

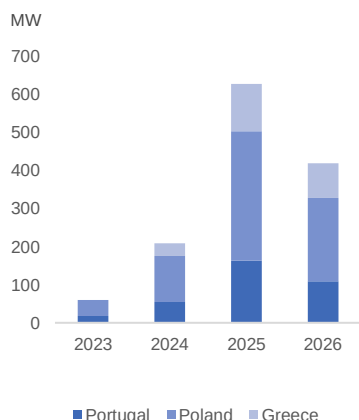


Figure 50: Method 1 - Develop to own - Pipeline Breakdown. Success rate 100%. Source: Analysts' estimates

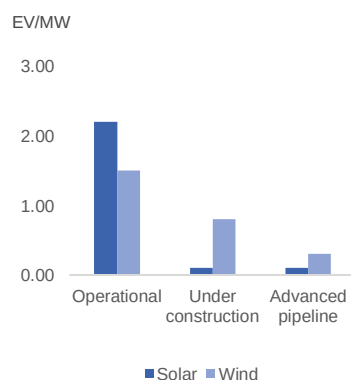


Figure 51: Method 2 - Deloitte - EV/MW Model

We notice that the **target** capital structure translates the **highly ambitious GreenVolt's CapEx plan for 2022e-2026e**, as some **moderate increase of financial debt is needed (as EV increases)**, and although current financing conditions may pose debt as expensive, we assume GreenVolt will continue to successfully be able to issue green bonds, since demand for these instruments are high.⁶⁵ Given GreenVolt's **past financial stability** (as reflected in the investment grade rating, with stable outlook), we also believe the Company is in a good path to sustain higher leverage levels with some **headroom**.⁶⁶ Notwithstanding, we stress that the **majority of CapEx funding sources'** will stem from **self-funding operations** (e.g., the biomass segment is currently the Company's cash cow) and from the **asset rotation strategy**.

As a result of using a **target capital structure**, the WACC is constant for all future years in our valuation.⁶⁷

EV/MW model

As mentioned earlier, in method 1, we derive value **based on the Company's strategy (develop to own vs. asset rotation)** and **on its estimates for CapEx and value creation** (the latter being, in turn, based on past V-Ridium transactions).

To incorporate in our model the **uncertainty** of developing and sell pipeline with such size, and where 96% is in early or advance stage (i.e., before licencing), we allocated **different mortality rates** (33%, 50%, 0%) to different scenarios (base, bear and bull, respectively) to the expected pipeline of 2025e and 2026e, for both develop to own/develop to sell.

For the assets to be sold as part of its **asset rotation strategy**, we split assets evenly (50:50), according to the stage they will reach (COD or RtB). Then, within each stage, we split between technology (solar PV and wind), according to the weight of each technology within total pipeline, excluding storage (63% and 37%, respectively). Using data from the Company, we allocated assets from each stage (UC, RtB, advanced and early stage) between 2023e-2026e, the last year for each we have visibility (Figure 49). Storage was considered to be sold at RtB. We applied a NPV until 2026e using WACC of 5.24%. These assets yield an **EV of €628m**. For the assets that the company will **keep on its balance sheet**, we considered the **remuneration schemes** in force in each country to reach **expected cash inflows**, and we considered **expected cash outflows to cover CapEx** (Figure 50). We applied a NPV until 2044e using the same WACC. These

⁶⁵ <https://www.bloomberg.com/news/articles/2022-10-04/green-bonds-defy-september-sales-slowdown-to-hit-four-month-high>

⁶⁶ <https://www.forbes.com/advisor/investing/investment-grade-bonds/>

⁶⁷ We assumed the same capital structure for both peer groups as we do not have evidence otherwise

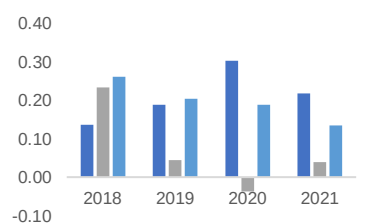


Figure 52: Profitability – Return on Sales – Gross Margin. Source: Bloomberg

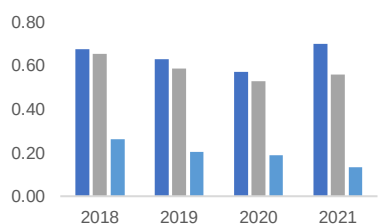


Figure 53: Profitability – Return on Sales – EBIT Margin. Source: Bloomberg

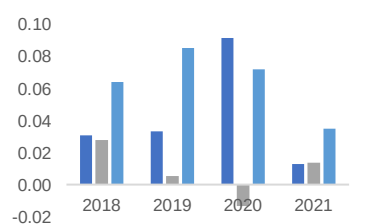


Figure 54: Profitability – Return on Investment – Return on Assets. Source: Bloomberg

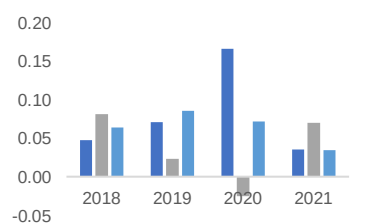


Figure 55: Profitability – Return on Investment – Return on Total Capital. Source: Bloomberg

assets yield an **EV of €163m**. Combining the two, we reach an **EV of €791m**, in the base scenario.

Having this method in mind, we wanted to **cross-check EV from this segment** (method 2), and we **derived value considering both the different stages of pipeline and precedent transactions**. We based our past transactions analysis on Deloitte's research papers concerning the valuation of solar PV and wind farm assets through a market approach, where the consulting firm ran **multiple regression analysis of transaction multiples**.^{68,69} These papers include **279 solar PV** and **327 wind farm** transactions from the past ten years.⁷⁰ The EV/MW coefficients are presented in Figure 51. This results in an **Enterprise Value (EV) of €1.032bn**, which is **€241m higher than the one achieved through method 1**, in the base case – which we find reasonable, since Deloitte's dataset includes a variety of transactions (in terms of stage and country).

Profitability, Liquidity, and Capital Structure

GreenVolt **gross margin ratio has been improving over the past four years**, a consequence of **increasing revenues which offset the nominal rise in costs**, indicating a **higher profitability** (Figure 52). The decrease in 2021a of EBIT and net income margin is related to the **new debt contracted that year**, which consequently increased interest rates (Figure 53). GreenVolt profitability related to sales is slightly above the peer groups. Looking at GreenVolt profitability related to investments, **Return on Assets ("ROA")** and **Return on Total Capital** peaked in 2020 at 0.09 and 0.17, respectively, both above peers. In 2021, it decreased to 0.01 and 0.04, respectively (Figure 54 and Figure 55). **The low ROA is explained by the fact that GreenVolt is an asset-heavy company and in a growth stage**. Therefore, its low ROA is acceptable.

To measure **liquidity**, we used the **current ratio** and the **cash ratio**. GreenVolt is significantly above the two peer groups with a currency ratio of 4.44 and a cash ratio of 3.50. Both ratios are above 1, which shows **GreenVolt healthy financial position to repay its obligations**. From 2020 to 2021, the improvement in both ratios is mainly due to the increase of €19m in cash equivalents. For this reason, GreenVolt surpasses both peer groups (Figure 56 and Figure 57). Moreover, in June 2022, GreenVolt **issued a capital increase of €100m, which strengthen its liquidity position**.

⁶⁸ <https://www2.deloitte.com/content/dam/Deloitte/nl/Documents/energy-resources/deloitte-nl-er-Valuing-onshore-wind-farm-assets-global-aug2017.pdf>

⁶⁹ https://www2.deloitte.com/content/dam/Deloitte/dk/Documents/pardot-downloads/Deloitte_Valuing_Solar_PV_Farm_Assets_Global_Mar2018.pdf

⁷⁰ Additionally, not only does Deloitte take into account the different stages of the farms' lifecycle, but it also has good regressions' R-square (0.87 and 0.89 for solar PV and wind, respectively). Therefore, we are comfortable applying Deloitte's conclusions to our EV/MW model

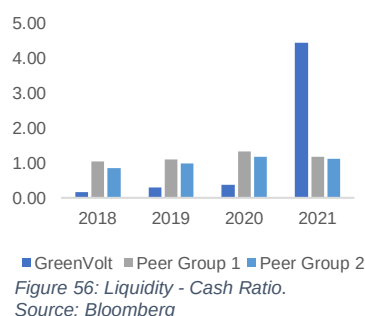


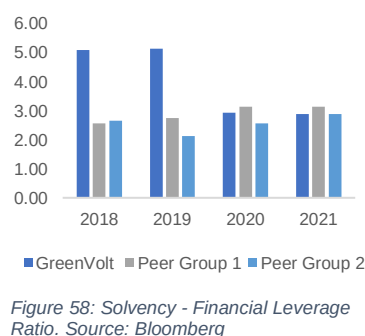
Figure 58 and Figure 59 show regularly used ratios to measure solvency. On average, **GreenVolt ratios have been decreasing over time and go in line with its peers**, which signals **low financial risk** and a **high probability that the firm will be able to support its capital-intensive projects** – which also supports our target capital structure, as mentioned earlier. Emphasis being placed on total debt to EBITDA ratio, which even though it has been decreasing, meaning GreenVolt would require less years to repay debt based on EBITDA, in 2021a it increased to values above its peers. Nevertheless, we expect it to decrease in the next years.

Terminal Value



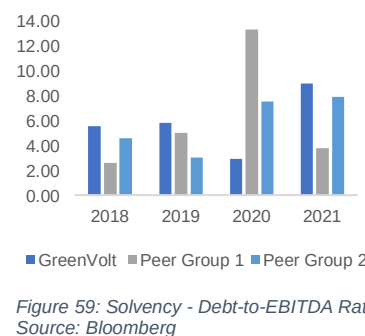
By 2044e, the growth rate is stabilized in all scenarios (i.e., **steady state**), and as such we applied a perpetuity, for which we used the **Gordon Growth Model** to calculate it. For **biomass**, looking at the base case, the level of NOPLAT used as a terminal cash flow is based on a normalized level of revenues around 1%, sustainable margins and Return on Invested Capital (“ROIC”) of c.5.8%, which reflects the midpoint of the firm business cycle. Same rationale applies for DG.

In almost all scenarios **ROIC is above WACC**, which means **GreenVolt is creating value** and it is expected to be sustained over time. For the biomass segment, given its moderate/low ROIC, GreenVolt should focus on improving returns before growing, while for distributed energy generation, given its high-ROIC level, GreenVolt should focus on growth (Figure 60 and Figure 61).



Overall, the **growth** results from the product of the reinvestment rate (“RR”) and ROIC after the firm reaches steady state in 2044e. **All growth rates are below the nominal GDP growth**, therefore we consider it to be a fair value as the Company will not be growing faster than the overall economy, as expected for this industry. The negative growth rate in the bear scenario for the biomass segment means the Company is shrinking and decaying in profit margins, which is partially by the fact that we do not factor in FiT extensions, which demonstrates their relevance for GreenVolt equity story.

Enterprise Value-to-Equity bridge and Target Price



Through the SOTP, we arrive at a **total EV of €1.605bn**, with a contribution of 33% (c.€534) from the biomass segment, 47% from the solar PV and wind (c.€753m), followed by 20% from distributed energy generation (c.€313m).

Then, we apply the following adjustments to EV: **net debt** (c.€199m), **provisions** (c.€16m), **non-controlling interests** (c.€53m), and **V-Ridium’s earnout**, which shareholders will receive in the short-time, subject to a certain level of success in

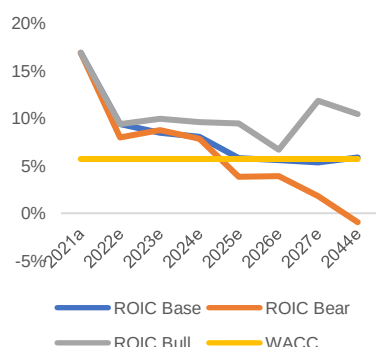


Figure 60: Biomass - ROIC per scenario and WACC 2021a-2027e and 2044. Source: Analysts' estimates

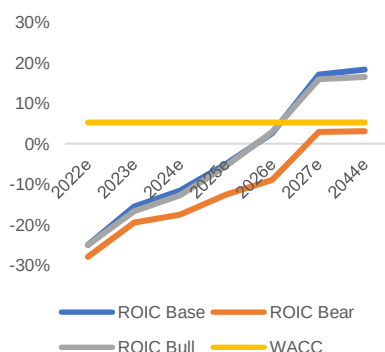


Figure 61: Distributed Generation - ROIC per scenario and WACC 2022e-2027e and 2044. Source: Analysts' estimates

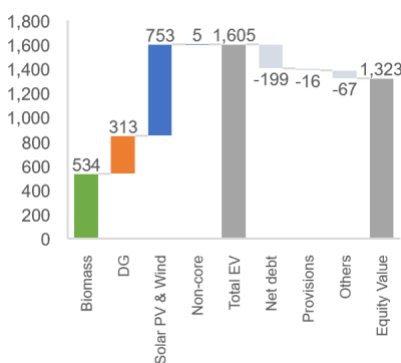


Figure 62: Enterprise Value to Equity Value bridge, in €m. Source: Analysts' estimates

		WACC				
		5.51%	5.61%	5.71%	5.81%	5.91%
Perpetuity growth rate	0.37%	543.9	535.0	526.4	518.2	510.4
	0.47%	548.0	538.8	530.1	521.7	513.6
	0.57%	552.2	542.8	533.8	525.2	517.0
	0.67%	556.6	547.0	537.8	528.9	520.5
	0.77%	561.2	551.3	541.8	532.8	524.1

Table 9: Growth vs WACC - Sensitivity analysis - Biomass (Base Scenario). Source: Analysts' estimates

the development strategy (c.€14m). We arrive at an **Equity Value ("EqV") of €1,323bn** (Figure 62).

With **c.139m of shares outstanding**, our EqV valuation implies a **Target Price ("TP") of €9.50**, which represents a **c.17% upside**, when compared to current market share price (€8.13).⁷¹ We arrive at this TP through a weighted average of the base scenario (50%), bear scenario (40%), and bull scenario (10%).

Sensitivity Analysis

To assess the **robustness** of our model underlyings, we will analyse the sensitiveness of each segment's EV against some factors we believe to have the most impact.

Biomass

First, we estimated the impact on EV from changing the segment's WACC (5.71%) and perpetuity growth rate (0.57%). Looking at Table 9, if we decrease WACC by 2pp and increase g by the same amount, EV increases by 5% (to c.€561m), and if we increase WACC by 2pp and decrease g by the same amount, EV decreases by 4% (to c.€510m). This means both variables are **robust**, since the variations they yield on EV are not considerably impactful. Then, since this segment is subject to some degree of uncertainty in the UK due to being **subject to electricity market prices**, we conducted a sensitivity analysis to the average selling price, holding g constant. As a starting point we have considered the lowest selling price forecasted (£199.74/MWh) and as a last point, we have considered the maximum selling price which coincides with the energy market cap (£285.61/MWh). In these extreme cases, the EV would decrease to c.€395m and increase to c.€596m, implying a downside and upside of 26% and 12%, respectively (Table 10). **This high sensitiveness indicates how crucial the average selling price is for GreenVolt, especially in the UK, since they are market-tied.**

Solar PV and wind

Assessing the impact on this segment's EV by changing WACC (5.24%) and the asset rotation strategy (75%), we conclude the percentage of assets to be sold at RtB or COD is a **resilient** factor, allowing GreenVolt some **headroom** to decide whether to farm down any given asset or to sell it. Regarding WACC, it depicts again our **high level of confidence** on this factor: a decrease of 2pp holding the asset rotation strategy level constant, leads to a small increase of 2% of the EV (Table 11).

⁷¹ As of December 14, 2022

		Average Selling Price (£/MWh)				
533.8		199.7	216.7	233.7	250.7	285.6
g	0.57%	395.7	435.4	475.2	515.0	596.5

Table 10: Growth vs Selling Price - Sensitivity analysis - Biomass (Base Scenario). Source: Analysts' estimates

		WACC				
752.8		5.04%	5.14%	5.24%	5.34%	5.44%
Asset rotation strategy	55.00%	760.7	747.5	734.4	721.6	709.0
	65.00%	765.0	754.2	743.6	733.1	722.8
	75.00%	769.3	761.0	752.8	744.6	736.7
	85.00%	773.6	767.7	761.9	756.2	750.5
	95.00%	777.9	774.5	771.1	767.7	764.4

Table 11: Asset strategy vs WACC - Sensitivity analysis - Solar and Wind. Analysts' estimates

		WACC				
313.4		5.04%	5.14%	5.24%	5.34%	5.44%
Perpetuity growth rate	1.80%	322.3	311.7	301.7	292.4	283.5
	1.90%	328.9	317.8	307.4	297.6	288.4
	2.00%	336.0	324.3	313.4	303.1	293.5
	2.10%	343.5	331.2	319.7	309.0	299.0
	2.20%	351.6	338.6	326.5	315.3	304.7

Table 12: Growth vs WACC - Sensitivity analysis - DG (Base Scenario). Source: Analysts' estimates

Trading multiples - Peer Group 1 - Biomass				
	EV / Sales	EV / EBITDA	EV / EBIT	EV / MW
Albioma	4.2x	11.5x	19.6x	2.9x
Drax	0.6x	5.5x	8.2x	1.4x
Ence	0.7x	2.6x	4.8x	2.5x
1st Quartile	0.6x	2.6x	4.8x	1.4x
Average	1.8x	6.5x	10.9x	2.3x
Median	0.7x	5.5x	8.2x	2.5x
3rd Quartile	4.2x	11.5x	19.6x	2.9x

Table 13: Trading multiples – Peer Group 1 (biomass). Source: Analysts' estimates

Trading multiples - Peer Group 2 - Solar PV and wind				
	EV / Sales	EV / EBITDA	EV / EBIT	EV / MW
Audax	0.3x	14.1x	24.0x	4.3x
EDP Renováveis	n.m.	12.2x	18.4x	1.9x
Voltaia	6.7x	15.0x	27.4x	2.8x
1st Quartile	n.m.	12.2x	18.4x	1.9x
Average	3.5x	13.8x	23.3x	3.0x
Median	3.5x	14.1x	24.0x	2.8x
3rd Quartile	n.m.	15.0x	27.4x	4.3x

Table 14: Trading multiples – Peer Group 2 (Solar PV and Wind). Source: Analysts' estimates

Distributed energy generation

As we have seen, WACC and perpetuity growth rate are resilient factors. If we hold g constant (c.2%), and we decrease WACC by 2pp, the EV of this segment increases by c.7% (Table 12). On the other hand, a change in the growth of the average selling price leads to greater changes in the EV: if the Company would be able to sustain its high-growth level of 10%, EV would increase to c.€817m (upside of 161%), and if the Company would not be able to grow its selling price (0%), EV would decrease to c.€228m (downside of 27%). **Selling price is usually a factor with low visibility**, more so in this segment, since we are basing our assumptions on a **benchmark analysis**. Installed capacity growth has also proven to be a highly sensitive factor in this segment, since the Company is currently in its ramp up phase and there is **little guidance on how this factor will evolve**. If GreenVolt could only increase installed capacity at the economy's level (c.2%), EV would decrease by 46% - which translates how important is the current growth phase in terms of this segment's value generation.

Multiples analysis

Trading multiples

In light of the Competitive Positioning chapter above, we believe **peer multiples are less relevant to derive GreenVolt's value as a whole**, since currently the Company is still between two very different realities. Nonetheless, we present the trading multiples for each Peer Group in Table 13 and Table 14. To capture GreenVolt's business mix, we will allocate 30% of weight to Peer Group 1 and the remainder to Peer Group 2, which goes in line with the Company's guidance on EBITDA split by 2026e, without considering solar and wind. Notwithstanding, a **first glimpse would suggest GreenVolt currently trades more closely to Peer Group 1 than the other**, since pure solar PV and wind players tend to have **higher market caps and higher revenues**, due to more diversified portfolios. Please refer to the combined trading multiples in Table 15. Having in mind the energy industry preferred multiples, we will allocate 50% weight to the EV/EBITDA and the remaining to EV/EBIT. We will disregard EV/Sales, since we believe it does not capture value well, and EV/MW, especially **due to the very early-stage pipeline of GreenVolt**. We arrive at implied TP of **€7.76**, using the average multiples mentioned earlier, which is lower than current share price. This suggests investors may have little visibility on the asset rotation/develop to own strategy, which we believe to ultimately be the **chief catalyst driving stock returns in the future** (as seen by the biggest contribution to EV, i.e., 47%).

Trading multiples - Combined				
	EV / Sales	EV / EBITDA	EV / EBIT	EV / MW
Average	3.0x	11.6x	19.5x	2.8x
Median	2.6x	11.5x	19.2x	2.7x

Table 15: Trading multiples – Peer Group 1 and Peer Group 2 combined. Source: Analysts' estimates.

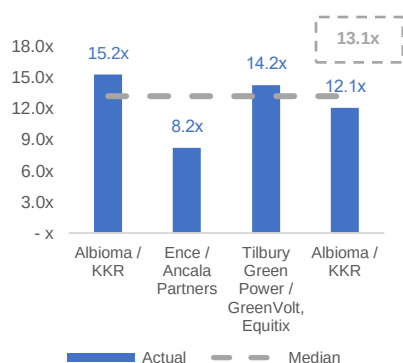


Figure 64: Precedent transactions multiples - EV / EBITDA – Peer Group 1 (biomass). Source: Analysts' estimates

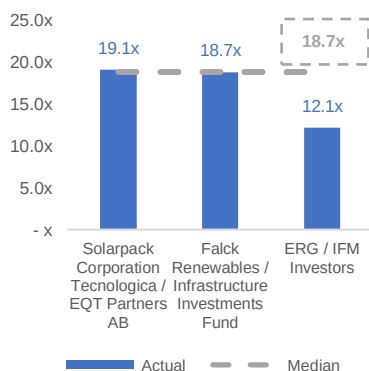


Figure 63: Precedent transactions multiples - EV / EBITDA – Peer Group 2 (Solar PV and Wind). Source: Analysts' estimates

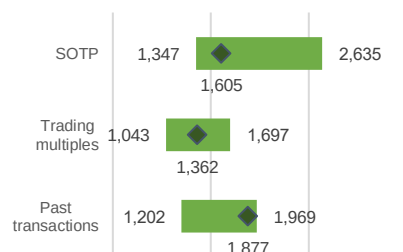


Figure 65: Football field. Source: Analysts' estimates

Precedent transactions

To find eligible transactions, we followed the rationale abovementioned, wherein we split the analysis according to different Peer Groups. We identified transactions within the last 2 years in Europe, with stakes acquired higher than 20% - please refer to Figure 64 and Figure 63. Then, we also allocated 30% to biomass transactions and 70% to solar PV and wind transactions. Again, we will only use EV/EBITDA to extrapolate EV. This way, we arrive at a TP of **€11.46**. **Precedent transactions yield a higher TP when compared to trading multiples and DCF (via SOTP)**, since **acquisition premiums** are factored in, as expected (Figure 65).

Recommendation

Although we recognise the **outstanding performance of the Tilbury acquisition in the UK** (which is expected to benefit from sky-rocketing wholesale prices in the upcoming years) and the **inflation-indexed biomass business in Portugal**, we find that the **geopolitical crisis in Europe** could negatively impact the **expected returns from the asset rotation strategy**, as buyers will face now greater uncertainty regarding construction costs.

Notwithstanding, we believe in GreenVolt's **ability to support future growth**. The core biomass business presents **high visibility of returns**, which enjoy a long-term fixed life. This, together with the **Company's expansion into solar PV and wind in Europe**, supports our equity growth story. Additionally, pursuing an asset rotation strategy for most of projects may help to curb risks while allowing **value creation at an early stage**, and **to finance future growth**. Regarding funding, GreenVolt raised €100m early in July to **prepare its balance sheet for daring investments** and to speed up its pipeline development. Moreover, in our view, the Company has capacity to take on **leverage** to step up its investment plan, to which high demand for green bonds is also favourable.

Nevertheless, **GreenVolt's equity story is largely based on early-stage pipeline**, to which a certain level of successful rate must be factored in (if the Company were to successfully execute 100% of the asset rotation pipeline, we would see c.42% upside), **and on management dependency** (CEO João Manso Neto and V-Ridium's deep expertise, being the former subject to a legal investigation). Since we believe GreenVolt has the necessary tools to carry on its ambitious asset developments, we view these **potential downsides as limited**. We initiate coverage at **Buy**.

Forecast Balance Sheet – Base Scenario

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Forecast Income Statement – Base Scenario

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2024
Total											
Revenue	50,537,103	64,283,355	89,877,619	130,709,839	156,291,156	168,612,781	171,966,542	166,239,855	170,221,151	165,674,619	192,756,706
Services rendered	-	-	-	8,313,674	32,040,334	57,118,860	89,260,636	119,851,809	134,559,710	137,250,904	192,184,401
Costs of sales	(19,870,281)	(24,880,975)	(39,028,957)	(43,215,283)	(57,089,860)	(70,353,239)	(86,222,191)	(105,213,224)	(113,145,671)	(97,671,105)	(130,598,489)
External supplies and services	(13,517,660)	(17,470,548)	(17,920,494)	(32,738,147)	(58,422,092)	(76,461,031)	(94,265,289)	(105,780,463)	(108,032,111)	(107,372,469)	(141,540,002)
Payroll expenses	-	-	-	(4,544,833)	(6,590,304)	(8,280,346)	(10,054,241)	(11,461,379)	(12,350,238)	(12,274,918)	(15,951,111)
Provisions and impairment reversals / (losses) in current assets	-	-	-	41	-	-	-	-	-	-	-
Amortization and depreciation	(7,764,671)	(10,623,246)	(12,148,457)	(25,807,060)	(26,787,303)	(27,900,663)	(29,114,409)	(31,430,725)	(31,578,318)	(31,611,637)	(32,291,777)
Impairment reversals / (losses) in non-current assets	(5,500,000)	-	6,335,742	-	-	-	-	-	-	-	-
Per Segment: Biomass and Others											
Revenue	50,537,103	64,283,355	89,877,619	130,708,543	154,790,854	165,938,167	167,786,876	160,627,744	163,920,337	162,929,601	188,913,018
% growth	-	27%	40%	45%	18%	7%	1%	-4%	2%	-1%	1%
Costs of sales	(19,870,281)	(24,880,975)	(39,028,957)	(41,507,026)	(43,673,606)	(45,951,739)	(47,634,528)	(53,152,106)	(54,875,357)	(55,672,328)	(71,790,062)
% growth	-	25%	57%	6%	5%	2%	4%	12%	3%	1%	2%
External supplies and services	(13,517,660)	(17,470,548)	(17,920,494)	(26,454,777)	(34,753,485)	(37,256,268)	(37,671,338)	(36,063,977)	(36,803,227)	(36,580,788)	(42,414,558)
As a % of revenue	27%	27%	20%	20%	22%	22%	22%	22%	22%	22%	22%
Payroll expenses	-	-	-	(4,120,863)	(4,880,109)	(5,231,551)	(5,289,836)	(5,064,129)	(5,167,935)	(5,136,700)	(5,955,882)
As a % of revenue	0%	0%	0%	3%	3%	3%	3%	3%	3%	3%	3%
Provisions and impairment reversals / (losses) in current assets	-	-	-	41	-	-	-	-	-	-	-
Amortization and depreciation	(7,764,671)	(10,623,246)	(12,148,457)	(25,584,852)	(26,352,398)	(27,142,969)	(27,957,259)	(29,912,312)	(29,912,312)	(29,912,312)	(29,912,312)
Impairment reversals / (losses) in non-current assets	(5,500,000)	-	6,335,742	-	-	-	-	-	-	-	-
Core operations result before taxes (Biomass and Others)	3,884,491	11,308,586	27,115,494	33,041,025	45,131,257	50,355,639	49,233,915	36,435,220	37,161,506	35,627,473	38,840,204
Per Segment: Distributed Generation											
Revenue	-	-	-	1,296	1,500,302	2,674,614	4,179,667	5,612,111	6,300,815	2,745,018	3,843,688
As a % of services rendered	-	-	-	0%	5%	5%	5%	5%	5%	2%	2%
Services rendered	-	-	-	8,313,674	32,040,334	57,118,860	89,260,636	119,851,809	134,559,710	137,250,904	192,184,401
% growth	-	-	-	0%	285%	78%	56%	34%	12%	2%	2%
Costs of sales	-	-	-	(1,708,257)	(13,416,254)	(24,401,501)	(38,587,663)	(52,061,118)	(58,270,314)	(41,998,777)	(58,808,427)
% growth	-	-	-	0%	685%	82%	58%	35%	12%	26%	26%
External supplies and services	-	-	-	(6,283,370)	(23,668,607)	(39,204,763)	(56,593,950)	(69,716,485)	(71,228,884)	(70,791,681)	(99,125,444)
As a % of total revenue	-	-	-	75%	71%	66%	61%	56%	51%	51%	51%
Payroll expenses	-	-	-	(423,970)	(1,710,195)	(3,048,795)	(4,764,405)	(6,397,250)	(7,182,303)	(7,138,218)	(9,995,229)
As a % of total revenue	-	-	-	5%	5%	5%	5%	5%	5%	5%	5%
Amortization and depreciation	-	-	-	(222,208)	(434,906)	(757,693)	(1,157,150)	(1,518,413)	(1,666,005)	(1,699,325)	(2,379,465)
Core operations result before taxes (distributed generation)	-	-	-	(322,835)	(5,689,326)	(7,619,277)	(7,662,866)	(4,229,347)	2,513,018	18,367,921	25,719,524
Core operations result before taxes (total)	3,884,491	11,308,586	27,115,494	32,718,190	39,441,930	42,736,362	41,571,050	32,205,873	39,674,524	53,995,394	64,559,728
Statutory taxes	815,743	2,374,803	5,694,254	6,870,820	(355,191)	8,974,636	8,729,920	6,763,233	8,331,650	11,339,033	13,557,543
Statutory tax rate	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%
Tax adjustments	(35,276)	473,312	1,075,114	483,993	-	-	-	-	-	-	-
Core operations result	3,104,024	8,460,471	20,346,126	25,363,377	39,797,121	33,761,726	32,841,129	25,442,640	31,342,874	42,656,361	51,002,185
Total											
Other income	3,313,368	851,448	222,437	643,758	583,400	583,400	583,400	583,400	583,400	583,400	583,400
Results related to investments	-	-	-	-	16,499	16,499	16,499	16,499	16,499	16,499	16,499
Other expenses	(364,828)	(82,425)	(129,539)	(126,507)	(120,792)	(120,792)	(120,792)	(120,792)	(120,792)	(120,792)	(120,792)
Per Segment: Biomass and Others											
Other income	3,313,368	851,448	222,437	627,479	567,121	567,121	567,121	567,121	567,121	567,121	567,121
Results related to investments	-	-	-	-	-	-	-	-	-	-	-
Other expenses	(364,828)	(82,425)	(129,539)	(114,554)	(108,839)	(108,839)	(108,839)	(108,839)	(108,839)	(108,839)	(108,839)
Non-core operations result before taxes (biomass and others)	2,948,540	769,023	92,898	512,925	458,282	458,282	458,282	458,282	458,282	458,282	458,282
Per Segment: Distributed Generation											
Other income	-	-	-	16,279	16,279	16,279	16,279	16,279	16,279	16,279	16,279
Results related to investments	-	-	-	16,499	16,499	16,499	16,499	16,499	16,499	16,499	16,499
Other expenses	-	-	-	(11,953)	(11,953)	(11,953)	(11,953)	(11,953)	(11,953)	(11,953)	(11,953)
Non-core operations result before taxes (distributed generation)	-	-	-	20,825	20,825	20,825	20,825	20,825	20,825	20,825	20,825
Non-core operations result before taxes (total)	2,948,540	769,023	92,898	533,750	479,107	479,107	479,107	479,107	479,107	479,107	479,107
Statutory taxes	619,193	161,495	19,509	112,088	100,612	100,612	100,612	100,612	100,612	100,612	100,612
Statutory tax rate	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%
Tax adjustments	(259,279)	-	-	3,240,232	-	-	-	-	-	-	-
Non-core operations result	2,588,626	607,528	73,389	(2,818,570)	378,495	378,495	378,495	378,495	378,495	378,495	378,495
Total											
Financial expenses	(620,739)	(1,872,466)	(1,791,223)	(8,951,635)	(20,252,070)	(14,346,849)	(14,923,371)	(15,916,754)	(16,392,398)	(16,761,471)	(13,591,774)
Financial income	443	480	67	700,807	-	-	-	-	-	-	-
Per Segment: Biomass and Others											
Financial expenses	(620,739)	(1,872,466)	(1,791,223)	(8,416,239)	(10,713,486)	(10,916,401)	(11,131,129)	(11,767,970)	(11,973,062)	(12,203,463)	(9,522,784)
As a % of prior-year total debt	-	-2%	-1%	9%	-4%	-4%	-4%	-4%	-4%	-4%	-4%
Financial income	443	480	67	658,891	-	-	-	-	-	-	-
As a % of prior-year excess cash	-	0%	0%	5%	5%	5%	5%	5%	5%	5%	5%
Financial result before taxes (biomass and others)	(620,296)	(1,871,986)	(1,791,156)	(7,757,348)	(10,713,486)	(10,916,401)	(11,131,129)	(11,767,970)	(11,973,062)	(12,203,463)	(9,522,784)
Per Segment: Distributed Generation											
Financial expenses	-	-	-	(535,396)	(3,116,753)	(3,430,449)	(3,792,242)	(4,148,784)	(4,419,336)	(4,558,008)	(4,068,990)
As a % of total debt	-	-	-	-2%	-2%	-2%	-2%	-2%	-2%	-2%	-2%
Financial income	-	-	-	41,915	-	-	-	-	-	-	-
As a % of excess cash	-	-	-	2%	0%	0%	0%	0%	0%	0%	0%
Financial result before taxes (distributed generation)	-	-	-	(493,480)	(3,116,753)	(3,430,449)	(3,792,242)	(4,148,784)	(4,419,336)	(4,558,008)	(4,068,990)
Financial result before taxes (total)	(620,296)	(1,871,986)	(1,791,156)	(8,250,828)	(20,252,070)	(14,346,849)	(14,923,371)	(15,916,754)	(16,392,398)	(16,761,471)	(13,591,774)
Statutory taxes	(130,262)	(393,117)	(376,143)	(1,732,674)	(4,252,935)	(3,012,838)	(3,133,908)	(3,342,518)	(3,442,404)	(3,519,909)	(2,854,273)
Statutory tax rate	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%
Non-controlling interests' share	-	(3,647)	(8,769)	(7,656,166)	(9,044,039)	(10,840,066)	(12,544,629)	(13,738,670)	(14,636,161)	(14,547,064)	(18,485,609)
As a % of revenues	-	-	-	5%	5%	5%	5%	5%	5%	5%	5%
Financial result	(490,034)	(1,475,222)	(1,406,244)	1,138,012	(6,955,097)	(493,945)	755,166	1,164,434	1,686,166	1,305,501	7,748,107
Total comprehensive income	5,202,616	7,592,777	19,013,271	23,682,819	33,220,519	33,646,275	33,974,790	26,985,569	33,407,535	44,340,357	59,128,787

EV/MW Model – Asset Rotation – Base Scenario

NPV - Asset rotation	2023	2024	2025	2026
RtB				
Solar	6,621	105,389	101,480	30,869
Wind	5,646	19,632	39,476	26,322
CoD				
Solar	11,918	41,441	83,329	55,564
Wind	12,288	42,728	85,918	57,290

CF	36,474	209,190	310,202	170,045
Discount CF	34,657	188,867	266,115	138,610

NPV	628
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EV/MW Model – Develop to Own – Base Scenario

NPV - Develop to own	2023	2024	2025	2026	2027	2044
MW	60	269	688	968	968	968
Portugal	18	72	181	254	254	254
Solar	18	72	181	254	254	254
Load factor	17%	17%	17%	17%	17%	17%
Energy generated (MWh)	27,121	108,017	272,192	382,140	382,140	382,140
PPA (€/MWh)	38	38	38	38	38	38
Expected cash inflow (€)	1,030,596	4,104,651	10,343,311	14,521,334	14,521,334	14,521,334
Expected cash outflow (€)	13,574,880	40,898,291	83,796,769	59,119,588	5,738,192	5,738,192
Poland	42	164	391	537	537	537
Solar	23	116	260	342	342	342
Load factor	17%	17%	17%	17%	17%	17%
Energy generated (MWh)	34,655	175,067	391,853	515,728	515,728	515,728
CFD (PLN/MWh)	340	340	340	340	340	340
Forward exchange rate (PL	0.21	0.21	0.21	0.21	0.21	0.21
CFD (€/MWh)	71	71	71	71	71	71
Expected cash inflow (€)	2,474,336	12,499,800	27,978,316	36,822,972	36,822,972	36,822,972
Expected cash outflow (€)	17,345,680	70,801,255	111,136,874	67,887,172	7,744,133	7,744,133
Wind	19	47	131	194	194	194
Load factor	39%	39%	39%	39%	39%	39%
Energy generated (MWh)	64,912	161,866	446,136	663,858	663,858	663,858
CFD (PLN/MWh)	230	230	230	230	230	230
CFD (€/MWh)	48	48	48	48	48	48
Expected cash inflow (€)	3,135,230	7,818,132	21,548,354	32,064,354	32,064,354	32,064,354
Expected cash outflow (€)	22,154,000	33,754,693	98,677,099	78,875,524	6,797,144	6,797,144
Greece	-	33	117	178	178	178
Solar	-	30	94	139	139	139
Load factor	17%	17%	17%	17%	17%	17%
Energy generated (MWh)	-	44,782	141,981	209,049	209,049	209,049
CFD (€/MWh)	54	54	54	54	54	54
Expected cash inflow (€)	-	2,418,218	7,666,975	11,288,662	11,288,662	11,288,662
Expected cash outflow (€)	-	25,162,292	63,693,289	48,503,327	4,018,831	4,018,831
Wind	-	4	23	39	39	39
Load factor	39%	39%	39%	39%	39%	39%
Energy generated (MWh)	-	12,447	77,169	132,846	132,846	132,846
CFD (€/MWh)	63	63	63	63	63	63
Expected cash inflow (€)	-	784,152	4,861,644	8,369,304	8,369,304	8,369,304
Expected cash outflow (€)	-	4,248,050	22,216,766	19,792,449	1,360,191	1,360,191
Romania	45	45	45	45	45	45
Solar	45	45	45	45	45	45
Load factor	17%	17%	17%	17%	17%	17%
Energy generated (MWh)	67,802	67,802	67,802	67,802	67,802	67,802
Market price (€/MWh)	37	37	37	37	37	37
Expected cash inflow (€)	2,508,689	2,508,689	2,508,689	2,508,689	2,508,689	2,508,689
Expected cash outflow (€)	1,018,116	1,018,116	1,018,116	1,018,116	1,018,116	1,018,116
Positive CF	9,148,851	30,133,642	74,907,289	105,575,314	105,575,314	105,575,314
Negative CF	54,092,676	175,882,696	380,538,913	275,196,176	26,676,607	26,676,607
CF	(44,943,825)	(145,749,054)	(305,631,623)	(169,620,862)	78,898,707	78,898,707
Discount CF	(42,704,926)	(131,589,623)	(262,193,630)	(138,264,619)	61,109,627	25,635,676

NPV	163
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Report Recommendations

Buy	Expected total return (including expected capital gains and expected dividend yield) of more than 10% over a 12-month period.
Hold	Expected total return (including expected capital gains and expected dividend yield) between 0% and 10% over a 12-month period.
Sell	Expected negative total return (including expected capital gains and expected dividend yield) over a 12-month period.

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